MATERIAL SAFETY DATA SHEETS

19 West Harrison Project
19 West Harrison
Seattle, WA 98119

One-Way Plumbing, LLC.

MASTER LIST OF CHEMICALS

TAB	CHEMICAL NAME	MANUFACTURER'S NAME
A	ABS BLACK CEMENT	OATEY
B	SILICONE SEALANT	POLYSEAMSEAL
C	PVC RAIN-R-SHINE BLUE CEMENT	OATEY
D	PVC PIPE FITTING COVERS & JACKETING	JOHNS MANVILLE
E	PVC MEDIUM CLEAR OR GRAY CEMENT	OATEY
F	PLUMBERS PUTTY	OATEY
G	PURPLE OR CLEAR PRIMER	OATEY
H	GREAT WHITE PIPE JOINT COMPOUND	OATEY
I	PVC HEAVY DUTY CLEAR OR GRAY CEMENT	OATEY
J	FLUX-RITE 90 PASTE	LA-CO INDUSTRIES
K	CPVC FLOWGUARD GOLD CEMENT 1-STEP	OATEY
L	DARK THREAD CUTTING OIL	OATEY
M	CAULK, TUB & TILE ADHEISIVE, WHITE	POLYSEAMSEAL
N	CAULK, TUB & TILE ADHEISIVE, CLEAR	POLYSEAMSEAL
O	ALL PURPOSE CEMENT	OATEY
P	STERLING LEAD-FREE SOLDER	LENOX
Q	FIRE STOP	3M
R		
S		
T		
U		
W		
X		
Y		
Z		

MATERIAL SAFETY DATA SHEET

MSDS Number: 1300E

Section 1 **PRODUCT AND COMPANY IDENTIFICATION**

Trade Name: OATEY ABS MEDIUM SPECIAL MILKY CLEAR or BLACK CEMENT
Product Nos.: Special Milky - 30922, 30923, 30924, 30929 Black - 30889, 30892, 30902, 30915, 30999, 32204, 32205, 32206, 32207
Product Use: Cement for ABS Plastic Pipe
Formula: ABS Resin in Solvent Solution
Synonyms: ABS Plastic Pipe Cement
Firm Name & Address: Oatey Company 4700 West 160th Street, Cleveland, Ohio 44135 www.oatey.com
Firm Phone No: (216) 267-7100
Emergency Phone Nos.: For Emergency First Aid call 1-877-740-5015. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared by: Technical Department
Preparation Date: 11/01/2009

Section 2 **HAZARDS IDENTIFICATION**

Emergency Overview:
Black or White
liquid with sharp, penetrating odor. Extremely flammable liquid and vapor. Vapors may cause flash fire. May cause eye and skin irritation. Inhalation of vapors or mist may cause respiratory irritation and central nervous system effects. Swallowing may cause irritation, nausea, vomiting, diarrhea and kidney or liver disorders. Aspiration hazard. May be fatal if swallowed. Symptoms may be delayed.

Section 3 **COMPOSITION/INFORMATION ON INGREDIENTS**

INGREDIENTS:	%wt/wt :	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA	OTHER:
Methyl Ethyl Ketone	40 – 60%	78-93-3	200 ppm 300 ppm	200 ppm	None
Acetone	10 - 20%	67-64-1	500 ppm 750 ppm STEL	1000 ppm	None
ABS Resin (Non-hazardous)	30 – 40%	9003-56-9	None Established	None Established	None

OSHA Hazard Classification: Flammable, irritant, organ effects

Section 4 **FIRST AID MEASURES**

Skin: Remove contaminated clothing immediately. Wash all exposed areas with soap and water. Get medical attention if irritation develops. Remove dried cement with Oatey Plumber's Hand Cleaner or baby oil.
Eyes: If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.
Inhalation: If symptoms of exposure develop, remove to fresh air. If breathing becomes difficult, administer oxygen. Administer artificial respiration if breathing has stopped. Seek immediate medical attention.
Ingestion: **DO NOT INDUCE VOMITING.** Rinse mouth with water. Never give anything by mouth to a person who is unconscious or drowsy. Get immediate medical attention by calling a Poison Control Center, or hospital emergency room. If medical advice

cannot be obtained, then take the person and product to the nearest medical emergency treatment center or hospital.

Section 5 FIRE FIGHTING MEASURES

Flashpoint / Method:	14 - 23 Degrees F. (-10 to -5 Degrees C) / CCCFP
Flammability:	LEL = 1.8 % Volume, UEL = 11.8 % Volume
Extinguishing Media:	Use dry chemical, CO2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure:	Firefighters should wear positive pressure self-contained breathing apparatus and full protective clothing for fires in areas where chemicals are used or stored
Unusual Fire And Explosion Hazards:	Extremely flammable liquid. Keep away from heat and all sources of ignition including sparks, flames, lighted cigarettes and pilot lights. Containers may rupture or explode in the heat of a fire. Vapors are heavier than air and may travel to a remote ignition source and flash back.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.

Section 6 ACCIDENTAL RELEASE MEASURES

Spill or Leak Procedures:	Remove all sources of ignition and ventilate area. Stop leak if it can be done without risk. Personnel cleaning up the spill should wear appropriate personal protective equipment, including respirators if vapor concentrations are high. Soak up spill with an inert absorbent such as sand, earth or other non-combusting material. Put absorbent material in covered, labeled metal containers. Prevent liquid from entering watercourses, sewers and natural waterways. Report releases to authorities as required. See Section 13 for disposal information.

Section 7 HANDLING AND STORAGE

Handling:	Avoid contact with eyes, skin and clothing. Avoid breathing vapors or mists. Use with adequate ventilation (equivalent to outdoors). Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep product away from heat, sparks, flames and all other sources of ignition. No smoking in storage or use areas. Keep containers closed when not in use.
Storage:	Store in a cool, dry, well-ventilated area away from incompatible materials. Keep containers closed when not in use.
Other:	"Empty" containers retain product residue and can be hazardous. Follow all MSDS precautions in handling empty containers. Do not cut or weld on or near empty or full containers.

Section 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Ventilation:	Open doors & windows. Provide ventilation capable of maintaining emissions at the point of use below recommended exposure limits. If used in enclosed area, use exhaust fans. Exhaust fans should be explosion-proof or set up in a way that flammable concentrations of solvent vapors are not exposed to electrical fixtures or hot surfaces.
Respiratory Protection:	For operations where the exposure limit may be exceeded, a NIOSH approved organic vapor respirator or supplied air respirator is recommended. Equipment selection depends on contaminant type and concentration, select in accordance with 29 CFR 1910.134 and good industrial hygiene practice. For firefighting, use self-contained breathing apparatus.
Skin Protection:	Rubber gloves are suitable for normal use of the product. For long exposures chemical resistant gloves may be required such as 4H(tm) or Silver Shield(tm) to avoid prolonged skin contact.
Eye Protection:	Safety glasses with side shields or safety goggles.

Section 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	151 Degrees F / 66 Degrees C
Melting Point:	Not applicable

Vapor Pressure:	145 mmHg @ 20 Degrees C
Vapor Density:	(Air = 1) 2.5
Volatile Components:	60-70%
Solubility In Water:	Negligible
pH:	Not applicable
Specific Gravity:	0.89 +/- 0.02 @ 20 Degrees C
Evaporation Rate:	(BUAC = 1) = 5.5 - 8.0
Appearance:	Yellow
Odor:	Sharp, penetrating odor
Will Dissolve In:	Methyl Ethyl Ketone
Material Is:	Liquid

Section 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, sparks, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.
Incompatibility/ Materials To Avoid:	Oxidizing agents, alkalis, amines, ammonia, acids, chlorine compounds, chlorinated inorganics (potassium, calcium and sodium hypochlorite) and hydrogen peroxides. May attack plastic, resins and rubber.
Hazardous Polymerization:	Will not occur.

Section 11 TOXICOLOGICAL INFORMATION

Inhalation:	Vapors or mists may cause mucous membrane and respiratory irritation, coughing, headache, dizziness, dullness, nausea, shortness of breath and vomiting. High concentrations may cause central nervous system depression, narcosis and unconsciousness. May cause kidney, liver and lung damage.
Skin:	May cause irritation with redness, itching and pain. Methyl ethyl ketone may be absorbed through the skin causing effects similar to those listed under inhalation.
Eye:	Vapors may cause irritation. Direct contact may cause irritation with redness, stinging and tearing of the eyes. May cause eye damage.
Ingestion:	Swallowing may cause abdominal pain, nausea, vomiting and diarrhea. Aspiration during swallowing or vomiting can cause chemical pneumonia and lung damage. May cause kidney and liver damage.
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis and damage to the kidney, liver, lungs and central nervous system.
Toxicity Data:	Acetone: Oral rat LD50: 5,800 mg/kg Inhalation rat LC50: 50,100 mg/m3/8 hours Methyl Ethyl Ketone: Oral rat LD50: 2,737 mg/kg Inhalation rat LC50: 23,500 mg/m3/8 hours Skin rabbit LD50: 6,480 mg/kg
Sensitization:	None of the components are known to cause sensitization.
Carcinogenicity:	None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA.
Mutagenicity:	Acetone, methyl ethyl ketone are generally thought not to be mutagenic.
Reproductive Toxicity:	Methyl ethyl ketone has been shown to cause embryofetal toxicity and birth defects in laboratory animals. Acetone has been found to cause adverse developmental effects only when exposure levels cause other toxic effects to the mother.
Medical Conditions Aggravated By Exposure:	Persons with pre-existing skin, lung, kidney or liver disorders may be at increased risk from exposure to this product.

Section 12 ECOLOGICAL INFORMATION

This product is not expected to be toxic to aquatic organisms.
Acetone: 96 hour LC50 for fish is greater than 100 mg/L.

	Methyl Ethyl Ketone: 96 hour LC50 for fish is greater than 100 mg/L.
VOC Information:	This product emits VOC's (volatile organic compounds) in its use. Make sure that use of this product complies with local VOC emission regulations, where they exist.
VOC Level:	Maximum 325 g/L per SCAQMD Test Method 316A.

Section 13 DISPOSAL CONSIDERATIONS

Waste Disposal:	Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number:	U002, U159
EPA Hazardous Waste ID Number:	D001, D035, F003, F0005
EPA Hazard Waste Number:	Ignitable Waste. Toxic Waste (Methyl Ethyl Ketone content)

Section 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)
UN/NA Number:	None	UN1133
Proper Shipping Name:	Consumer Commodity	Adhesives
Hazard Class:	ORM-D	3
Packing Group:	None	PGII
Hazard Labels:	None	Flammable Liquid
IMDG		
UN Number:	UN1133	UN1133
Proper Shipping Name:	Adhesives	Adhesives
Hazard Class:	3	3
Packing Group:	II	II
Label:	None (Limited Quantities are expected from labeling)	Class 3 (Flammable Liquid)
Flashpoint (deg C)	-10 to -5 Degrees C	-10 to -5 Degrees C

2008 North American Emercency Response Guidebook Number: 127

Section 15 REGULATORY INFORMATION

Hazard Category for Section 311/312:	Acute Health, Chronic Health, Flammable
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain chemicals subject to SARA Title III Section 313 Reporting requirements.
CERCLA 103 Reportable Quantity:	Spills of this product over the RQ (reportable quantity) must be reported to the National Response Center. The RQ for the product, based on the RQ for Methyl Ethyl Ketone (60% maximum) of 5,000 lbs, is 8,333 lbs. Many states have more stringent release reporting requirements. Report spills required under federal, state and local regulations.
California Proposition 65:	This product does not contain any chemicals subject to California Proposition 65 regulations.
TSCA Inventory	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Class B, Division 2; Class D, Division 2, Subdivision B; Class D, Division 2, Subdivision A. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

Section 16 OTHER INFORMATION

NFPA and HMIS:
NFPA Hazard Signal: Health: 2 Flammability: 3 Reactivity: 1 Special: None
HMIS Hazard Signal: Health: 2* Flammability: 3 Reactivity: 1 PPE: G

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, we cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

Template: tmpl-us-e3

MATERIAL SAFETY DATA SHEET

MSDS Name: POLYSEAMSEAL 100% SILICONE SEALANT - SQUEEZE TUBE
MSDS Number: sD11526
Version Number
MSDS Date: MAY-28-1999
Page Number: 1

SECTION I - PRODUCT AND COMPANY INFORMATION

Product Name:	POLYSEAMSEAL 100% SILICONE SEALANT - SQUEEZE TUBE
Hazard Rating:	Health: 1 Fire: 1 Reactivity: 0 PPI: B
Company Identification:	OSI SEALANTS, INC. 7405 PRODUCTION DRIVE MENTOR OH 44060
Contact:	Safety Officer
Telephone/Fax:	(440) 255-8900 (440) 974-2395
Emergency Phone (24 hour)	CHEMTREC (800) 424-9300
Chemtrec (outside-USA)	(703) 527-3887
Preparer	T.F.Barr Sr. R.&D. Chemist
Product Class	SEALANT
Trade Name	POLYSEAMSEAL
Product Code	Squeeze tube

Description: Low modulus, neutral-cure silicone sealant.

SECTION II - INGREDIENT AND HAZARD INFORMATION

Ingredient Name	CAS Number	Percent	TSCA
METHYL ETHYL KETOXIME ***	96-29-7	< 5.0	Y

Ingredient Notes:
***MEKO is not a component but is released upon curing
in the presence of humid air.

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SECTION III - PHYSICAL AND CHEMICAL PROPERTIES

Form:	Non-sag, high viscosity caulk.
Appearance/Color:	White, Clear
Odor:	Mild ammonia-like
Solubility (in water):	insol.
pH Value, +/-.3:	Not Applicable

Boiling Range:	Not Applicable
Vapor Pressure (mmHg):	1.@ 68.øF (20.øC)
Evaporation Rate:	0.5 times Slower than n-Butyl Acetate
Vapor Density:	Heavier than air
% Volatile, Weight	2.%
% Volatile, Volume	2.4%
Specific Gravity:	1.06
VOC (less H2O or exempt)	< 5 g/l
Heavy Elements (ppm)	0.

NOTE:
Odor : Mild ammonia-like odor. pH : N/A
Physical State : Semi-solid Freeze Point : N/A

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SECTION IV - FIRE FIGHTING MEASURES (Flash,UEL,LEL for solvent only)

Flammability Class	III B
Flash Range:	over 212.øF (100.øC) Tag Closed Cup
Explosive Range (LEL/UEL):	Not Applicable

EXTINGUISHING MEDIA:
Water---Carbon Dioxide---Dry Chemical---Foam
use water to cool material in vicinity of fire.
SPECIAL FIRE-FIGHTING PROCEDURES:
Use self-contained breathing apparatus with full facepiece
operated in pressure demand or other positive pressure mode.
UNUSUAL FIRE AND EXPLOSION HAZARDS:
Water may be unsuitable except as a cooling agent.
Use fog nozzle. Any container may rupture when exposed to
extreme heat.

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SECTION V - HEALTH HAZARD DATA

ROUTES OF ENTRY:
ENTRY THROUGH...
Inhalation? Yes | Skin? Yes | Ingestion? Yes
TARGET ORGANS...
Note: Methylethyl Ketoxime (MEKO) is formed upon contact

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with water or humid air.
Normal Exposure -- Methyl Ethyl Ketoxime: Skin, Eye
Overexposure -- Methyl Ethyl Ketoxime: Male rodents exposed to MEKO vapors over their lifetime developed liver cancer. Additional testing is planned to determine any relevance to humans.

CARCINOGENICITY...
NTP? N/E | IARC Monographs? N/E | OHSA? NO

EFFECTS OF OVEREXPOSURE
Eyes: Direct contact can cause irritation, redness and tearing.
Skin: Prolonged or repeated contact can cause moderate irritation. Oximes are possible skin sensitizers.
Inhalation: Excessive inhalation of vapors can cause respiratory irritation, headache, and drowsiness. High concentrations of vapors are anesthetic and may have blood oxygen effects.
Ingestion: Small amounts transferred to the mouth during use will not injure. Large amounts could injure slightly.

FIRST AID MEASURES
Eyes: Flush with large amounts of water for at least 15 minutes lifting eyelids occasionally; get prompt medical attention
Skin: Wash thoroughly with soap and water; apply a mild skin cream.
Inhalation: Remove affected person to fresh air; if breathing is difficult, administer oxygen, get medical attention.
Ingestion: DO NOT INDUCE VOMITING! Get immediate medical attention.
Cured sealant is non-hazardous.

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SECTION VI - STABILITY AND REACTIVITY
Stability: This product is stable
Hazardous Polymerization: Hazardous polymerization will not occur

INCOMPATABILITY:
Strong oxidizing agents cause a reaction. Air and moisture cause curing and release of Methylethyl Ketoxime.

CONDITIONS TO AVOID:
Exposure to air and moisture untill ready to use.

HAZARDOUS DECOMPOSITION PRODUCTS:
Silicone dioxide, carbon dioxide and traces of incompletely burned carbon products.

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SECTION VII - ACCIDENTAL RELEASE AND DISPOSAL MEASURES:

STEPS TO BE TAKEN IN CASE OF SPILL:
Ventilate confined areas. (Open windows and doors)
Scoop into sealable safety container.
Clean spill area to reduce risk of slipping hazard.
WASTE DISPOSAL METHOD:
Dispose of following local, state, and federal regulations.
Cured material is not a hazardous waste.

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SECTION VIII - EXPOSURE CONTROLS/PERSONAL PROTECTION

Occupational Exposure Limits

	ACGIH TLV	ACGIH TLV-C	ACGIH STEL	OSHA STEL	OSHA PEL
METHYL ETHYL KETOXIME ***	10.00 PPM	N/est	10.00 PPM	N/est	N/est

RESPIRATORY PROTECTION:
Use NIOSH approved respiratory protection if TLV is exceeded.
VENTILATION:
Use only in well ventilated areas.
Provide sufficient mechanical (general and/or local exhaust) ventilation to maintain exposure below TLV.
PROTECTIVE CLOTHING:
Rubber gloves should be worn.
EYE PROTECTION:
Chemical goggles should be worn.
HANDLING AND STORAGE PRECAUTIONS:
Keep container closed when not in use.
Avoid contact with eyes and skin. Do not handle contact lenses until all sealant has been removed from hands. Residual sealant may transfer to lenses and cause severe eye irritation.
Keep out of reach of children.

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SECTION IX - TRANSPORT INFORMATION:

DOT CLASSIFICATIONS FOR CARTRIDGES 10 or 29 oz.

DESCRIPTION: Non-hazardous
DOT CLASSIFICATIONS for 1 GALLON or larger CONTAINERS

DESCRIPTION: Non-hazardous
Caulking type product

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SECTION X - REGULATORY INFORMATION:

The following ingredients are registered for TSCA 12B

Ingredient Name	CAS Number	Percent
METHYL ETHYL KETOXIME ***	96-29-7	2.40

This material fits the EPA Hazard Category definition of Immediate (Acute) and Delayed (Chronic) Health Hazards under SARA Sections 311, 312.
SARA Section 313 toxic chemicals: None

Chemical Substances have been reported to the EPA Office of Toxic Substances in accordance with the requirements of the Toxic Chemical Substances Control Act (Title 40 CFR 710).
California PROP.65 Chemicals: none known

DISCLAIMER:
The information contained herein is based on data available as of the date of preparation of this MSDS and which we believe to be reliable. However, no warranty is expressed or implied regarding the accuracy of the data. We shall not be responsible for the use of this information, or of any product, method, or apparatus mentioned, and the user must make his own investigation to determine the suitability of the information or products for his particular purpose, for the protection of the environment, and the health and safety of the users of this material.

MATERIAL SAFETY DATA SHEET

MSDS Number: 1104E

Section 1 **PRODUCT AND COMPANY IDENTIFICATION**

Trade Name: OATEY PVC RAIN-R-SHINE BLUE CEMENT
Product Nos.: 30890, 30891, 30893, 30894, 30895, 30896, 31954, 31955, 31956, 31957
Product Use: Cement for PVC Plastic Pipe
Formula: PVC Resin in Solvent Solution
Synonyms: PVC Plastic Pipe Cement
Firm Name & Address: Oatey Company 4700 West 160th Street, Cleveland, Ohio 44135 www.oatey.com
Firm Phone No: (216) 267-7100
Emergency Phone Nos.: For Emergency First Aid call 1-877-740-5015. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared by: Technical Department
Preparation Date: 10/01/2009

Section 2 **HAZARDS IDENTIFICATION**

Emergency Overview:
Blue
liquid with an ether-like odor. Extremely flammable liquid and vapor. Vapors may cause flash fire. May cause eye and skin irritation. Inhalation of vapors or mist may cause respiratory irritation and central nervous system effects. Swallowing may cause irritation, nausea, vomiting, diarrhea and kidney or liver disorders. Aspiration hazard. May be fatal if swallowed. Symptoms may be delayed.

Section 3 **COMPOSITION/INFORMATION ON INGREDIENTS**

INGREDIENTS:	%wt/wt :	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA	OTHER:
Tetrahydrofuran	30 - 60%	109-99-9	50 ppm(skin) 100 ppm STEL	200 ppm	25 ppm (Mfg)
Methyl Ethyl Ketone	10 – 30%	78-93-3	200 ppm 300 ppm	200 ppm	None
Acetone	10 - 30%	67-64-1	500 ppm 750 ppm STEL	1000 ppm	None
PVC Resin (Non-hazardous)	12 – 20%	9002-86-2	10 mg/m3	15 mg/m3	None
Cyclohexanone	7 – 13%	108-94-1	20 ppm(skin) 50 ppm STEL	50 ppm	None
Amorphous Fumed Silica (Non-hazardous)	1 - 5%	112945-52-5	10 mg/m3	None Established	None a

OSHA Hazard Classification: Flammable, irritant, organ effects

Section 4 **FIRST AID MEASURES**

Skin: Remove contaminated clothing immediately. Wash all exposed areas with soap and water. Get medical attention if irritation develops. Remove dried cement with hand cleaner or baby oil.

Eyes:	If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.
Inhalation:	If symptoms of exposure develop, remove to fresh air. If breathing becomes difficult, administer oxygen. Administer artificial respiration if breathing has stopped. Seek immediate medical attention.
Ingestion:	**DO NOT INDUCE VOMITING.** Rinse mouth with water. Never give anything by mouth to a person who is unconscious or drowsy. Get immediate medical attention by calling a Poison Control Center, or hospital emergency room. If medical advice cannot be obtained, then take the person and product to the nearest medical emergency treatment center or hospital.

Section 5 FIRE FIGHTING MEASURES

Flashpoint / Method:	14 - 23 Degrees F. (-10 to -5 Degrees C) / CCCFP
Flammability:	LEL = 1.8 % Volume, UEL = 11.8 % Volume
Extinguishing Media:	Use dry chemical, CO_2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure:	Firefighters should wear positive pressure self-contained breathing apparatus and full protective clothing for fires in areas where chemicals are used or stored
Unusual Fire And Explosion Hazards:	Extremely flammable liquid. Keep away from heat and all sources of ignition including sparks, flames, lighted cigarettes and pilot lights. Containers may rupture or explode in the heat of a fire. Vapors are heavier than air and may travel to a remote ignition source and flash back. This product contains tetrahydrofuran that may form explosive organic peroxide when exposed to air or light or with age.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.

Section 6 ACCIDENTAL RELEASE MEASURES

Spill or Leak Procedures:	Remove all sources of ignition and ventilate area. Stop leak if it can be done without risk. Personnel cleaning up the spill should wear appropriate personal protective equipment, including respirators if vapor concentrations are high. Soak up spill with an inert absorbent such as sand, earth or other non-combusting material. Put absorbent material in covered, labeled metal containers. Prevent liquid from entering watercourses, sewers and natural waterways. Report releases to authorities as required. See Section 13 for disposal information.

Section 7 HANDLING AND STORAGE

Handling:	Avoid contact with eyes, skin and clothing. Avoid breathing vapors or mists. Use with adequate ventilation (equivalent to outdoors). Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep product away from heat, sparks, flames and all other sources of ignition. No smoking in storage or use areas. Keep containers closed when not in use.
Storage:	Store in a cool, dry, well-ventilated area away from incompatible materials. Keep containers closed when not in use.
Other:	"Empty" containers retain product residue and can be hazardous. Follow all MSDS precautions in handling empty containers. Do not cut or weld on or near empty or full containers.

Section 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Ventilation:	Open doors & windows. Provide ventilation capable of maintaining emissions at the point of use below recommended exposure limits. If used in enclosed area, use exhaust fans. Exhaust fans should be explosion-proof or set up in a way that flammable concentrations of solvent vapors are not exposed to electrical fixtures or hot surfaces.

Respiratory Protection:	For operations where the exposure limit may be exceeded, a NIOSH approved organic vapor respirator or supplied air respirator is recommended. Equipment selection depends on contaminant type and concentration, select in accordance with 29 CFR 1910.134 and good industrial hygiene practice. For firefighting, use self-contained breathing apparatus.
Skin Protection:	Rubber gloves are suitable for normal use of the product. For long exposures chemical resistant gloves may be required such as 4H(tm) or Silver Shield(tm) to avoid prolonged skin contact.
Eye Protection:	Safety glasses with side shields or safety goggles.

Section 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	151 Degrees F / 66 Degrees C
Melting Point:	Not applicable
Vapor Pressure:	145 mmHg @ 20 Degrees C
Vapor Density:	(Air = 1) 2.5
Volatile Components:	80-84%
Solubility In Water:	Negligible
pH:	Not applicable
Specific Gravity:	0.94 +/- 0.02 @ 20 Degrees C
Evaporation Rate:	(BUAC = 1) = 5.5 - 8.0
Appearance:	Blue Liquid
Odor:	Ether-Like
Will Dissolve In:	Tetrahydrofuran
Material Is:	Liquid

Section 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, sparks, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.
Incompatibility/ Materials To Avoid:	Oxidizing agents, alkalis, amines, ammonia, acids, chlorine compounds, chlorinated inorganics (potassium, calcium and sodium hypochlorite) and hydrogen peroxides. May attack plastic, resins and rubber.
Hazardous Polymerization:	Will not occur.

Section 11 TOXICOLOGICAL INFORMATION

Inhalation:	Vapors or mists may cause mucous membrane and respiratory irritation, coughing, headache, dizziness, dullness, nausea, shortness of breath and vomiting. High concentrations may cause central nervous system depression, narcosis and unconsciousness. May cause kidney, liver and lung damage.	
Skin:	May cause irritation with redness, itching and pain. Methyl ethyl ketone and cyclohexanone may be absorbed through the skin causing effects similar to those listed under inhalation.	
Eye:	Vapors may cause irritation. Direct contact may cause irritation with redness, stinging and tearing of the eyes. May cause eye damage.	
Ingestion:	Swallowing may cause abdominal pain, nausea, vomiting and diarrhea. Aspiration during swallowing or vomiting can cause chemical pneumonia and lung damage. May cause kidney and liver damage.	
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis and damage to the kidney, liver, lungs and central nervous system.	
Toxicity Data:	Acetone:	Oral rat LD50: 5,800 mg/kg Inhalation rat LC50: 50,100 mg/m3/8 hours

Cyclohexanone:	Oral rat LD50: 1,620 mg/kg Inhalation rat LC50: 8,000 ppm/4 hours Skin rabbit LD50: 1 mL/kg
Tetrahydrofuran:	Oral rat LD50: 1,650 mg/kg Inhalation rat LC50: 21,000 ppm/3 hours
Methyl Ethyl Ketone:	Oral rat LD50: 2,737 mg/kg Inhalation rat LC50: 23,500 mg/m3/8 hours Skin rabbit LD50: 6,480 mg/kg

Sensitization: None of the components are known to cause sensitization.

Carcinogenicity: None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA. The National Toxicology Program has reported that exposure of mice and rats to tetrahydrofuran (THF) vapor levels up to 1800 ppm 6 hr/day, 5 days/week for their lifetime caused an increased incidence of kidney tumors in male rats and liver tumors in female mice. The significance of these findings for human health is unclear at this time, and may be related to "species specific" effects. Elevated incidences of tumors in humans have not been reported for THF. ACGIH has classified cyclohexanone (CYH) and tetrahydrofuran as “A3,” Confirmed Animal Carcinogens with Unknown Relevance to Humans.

Mutagenicity: Cyclohexanone has been positive in bacterial and mammalian assays. Acetone, methyl ethyl ketone and tetrahydrofuran are generally thought not to be mutagenic.

Reproductive Toxicity: Methyl ethyl ketone and cyclohexanone have been shown to cause embryofetal toxicity and birth defects in laboratory animals. Acetone and tetrahydrofuran has been found to cause adverse developmental effects only when exposure levels cause other toxic effects to the mother.

Medical Conditions Aggravated By Exposure: Persons with pre-existing skin, lung, kidney or liver disorders may be at increased risk from exposure to this product.

Section 12 ECOLOGICAL INFORMATION

This product is not expected to be toxic to aquatic organisms.
Cyclohexanone: 96 hour LC50 values for fish is over 100 mg/l.
Tetrahydrofuran: 96 hour LC50 fathead minnow: 2160 mg/L.
Acetone: 96 hour LC50 for fish is greater than 100 mg/L.
Methyl Ethyl Ketone: 96 hour LC50 for fish is greater than 100 mg/L.

VOC Information: This product emits VOC’s (volatile organic compounds) in its use. Make sure that use of this product complies with local VOC emission regulations, where they exist.

VOC Level: Maximum 510 g/L per SCAQMD Test Method 316A.

Section 13 DISPOSAL CONSIDERATIONS

Waste Disposal:	Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number:	U002, U057, U159, U213
EPA Hazardous Waste ID Number:	D001, D035, F003, F0005
EPA Hazard Waste Number:	Ignitable Waste. Toxic Waste (Methyl Ethyl Ketone content)

Section 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)
UN/NA Number:	None	UN1133
Proper Shipping Name:	Consumer Commodity	Adhesives

Hazard Class:	ORM-D	3
Packing Group:	None	PGII
Hazard Labels:	None	Flammable Liquid
IMDG		
UN Number:	UN1133	UN1133
Proper Shipping Name:	Adhesives	Adhesives
Hazard Class:	3	3
Packing Group:	II	II
Label:	None (Limited Quantities are expected from labeling)	Class 3 (Flammable Liquid)
Flashpoint (deg C)	-10 to -5 Degrees C	-10 to -5 Degrees C

2008 North American Emercency Response Guidebook Number: 127

Section 15 REGULATORY INFORMATION

Hazard Category for Section 311/312:	Acute Health, Chronic Health, Flammable
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain chemicals subject to SARA Title III Section 313 Reporting requirements.
CERCLA 103 Reportable Quantity:	Spills of this product over the RQ (reportable quantity) must be reported to the National Response Center. The RQ for the product, based on the RQ for Tetrahydrofuran (60% maximum) of 1,000 lbs, is 1,667 lbs. Many states have more stringent release reporting requirements. Report spills required under federal, state and local regulations.
California Proposition 65:	This product contains trace amounts of chemicals known to the State of California to cause cancer. Under normal use conditions, exposure to these chemicals at levels above the State of California “No Significant Risk Level” (NSRL) are unlikely. The use of proper personal protective equipment (PPE) and ventilation guidelines noted in Section 8 will minimize exposure to these chemicals.
TSCA Inventory	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Class B, Division 2; Class D, Division 2, Subdivision B; Class D, Division 2, Subdivision A. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

Section 16 OTHER INFORMATION

NFPA and HMIS:	
NFPA Hazard Signal:	Health: 2 Flammability: 3 Reactivity: 1 Special: None
HMIS Hazard Signal:	Health: 2* Flammability: 3 Reactivity: 1 PPE: G

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, we cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

Template: tmpl-us-e1

Section 1 - Product and Company Identification

Hazard Label No label required

Company Information

Johns Manville
Insulation Systems
P.O. Box 5108
Denver, CO 80127 USA

Telephone: 303-978-2000 8:00AM-5:00PM M-F
Internet Address: http://www.jm.com
Emergency: 800-424-9300 (Chemtrec, In English)

Trade Names:

Ceel-Co® 300 PVC Fitting Covers and Jacketing;
Ceel-Co® 550 PVC Fitting Covers and Jacketing;
PVC Drain Pans;
Zeston® 300 Series PVC Fitting Covers and Jacketing;
Zeston® 2000 Series PVC Fitting Covers and Jacketing;
Z-Tape

Use: Specifically designed for industrial and commercial applications.

Section 3 - Composition/Information on Ingredients

General Product Description

Pre-formed, molded plastic sheets and shapes of varying colors.

Section 4 - First Aid Measures

First Aid: Inhalation

None required

First Aid: Skin

None required

First Aid: Ingestion

None required

First Aid: Eyes

None required

First Aid: Ears

None required

Section 5 - Fire Fighting Measures

Flash Point: Not applicable
Method Used: Not applicable
Upper Flammable Limit (UFL): Not applicable
Lower Flammable Limit (LFL): Not applicable
Auto Ignition: Not determined
Flammability Classification: Not determined
Rate of Burning: Not determined

General Fire Hazards

There is no potential for spontaneous fire or explosion. If exposed to fire, product may melt and drops of molten product may cause burns.

Extinguishing Media

Carbon dioxide (CO_2), dry chemical.

Fire Fighting Equipment/Instructions

No special procedures are expected to be necessary for this product. Normal fire fighting procedures should be followed to avoid inhalation of smoke and gases.

Section 6 - Accidental Release Measures

Clean-Up Procedures

Pick up large pieces. Vacuum dusts.

Section 7 - Handling and Storage

Handling Procedures

Use protective equipment as described in Section 8 of this safety data sheet when handling uncontained material. Handle in accordance with good industrial hygiene and safety practices.

Storage Procedures

Warehouse storage should be in accordance with package directions, if any. Material should be kept clean, dry, and in original packaging.

Section 8 - Exposure Controls / Personal Protection

A: Component Exposure Limits

This material has no components listed.

PERSONAL PROTECTIVE EQUIPMENT

Personal Protective Equipment: Eyes/Face

None required

Personal Protective Equipment: Ears

None required

Personal Protective Equipment: Skin

None required

Personal Protective Equipment: Respiratory

None required

Ventilation

None required

Personal Protective Equipment: General

None required

Section 9 - Physical & Chemical Properties

Appearance:	Pre-formed, molded plastic sheets and shapes of varying colors.	**Odor:**	No odor
Physical State:	Solid	**pH:**	Not determined
Vapor Pressure:	Not determined	**Vapor Density:**	Not determined
Boiling Point:	Not determined	**Melting Point:**	Not determined
Solubility (H_2O):	Nil	**Specific Gravity:**	Variable
Freezing Point:	Not applicable	**Evaporation Rate:**	Not applicable
Percent Volatile:	0	**VOC:**	Not determined

Section 10 - Stability & Reactivity Information

Stability

These products are not reactive.

Hazardous Decomposition

This product will decompose at temperatures >300℃. May form carbon dioxide and carbon monoxide.

Hazardous Polymerization

Will not occur.

Section 11 - Toxicological Information

Acute Toxicity

A: General Product Information

If dust evolves from this product during use it may cause temporary mechanical irritation or scratchiness of the throat and/or itching of the eyes and skin.

B: Component Analysis - LD50/LC50

This material has no components listed.

Component Carcinogenicity

This material has no components listed.

Section 12 - Ecological Information

Ecotoxicity

A: General Product Information

No data available for this product.

B: Component Analysis - Ecotoxicity - Aquatic Toxicity

This material has no components listed.

Section 13 - Disposal Considerations

US EPA Waste Number & Descriptions

A: General Product Information

This product is not expected to be a hazardous waste when it is disposed of according to the U.S. Environmental Protection Agency (EPA) under Resource Conservation and Recovery Act (RCRA) regulations. Product characterization after use is recommended to ensure proper disposal under federal and/or state requirements.

B: Component Waste Numbers

This material has no components listed.

Disposal Instructions

Dispose of waste material according to Local, State, Federal, and Provincial Environmental Regulations.

Section 14 - Transport Information

International Transport Regulations

These products are not classified as dangerous goods according to international transport regulations.

Section 15 - Regulatory Information

US Federal Regulations

A: General Product Information

SARA 311/312: This product is not classified as hazardous under SARA 311/312.

B: Component Analysis

This material has no components listed.

State Regulations

A: General Product Information

Other state regulations may apply. Check individual state requirements.

B: Component Analysis - State

This material has no components listed.

A: TSCA Status

This material has no components listed.

International Regulations

A: General Product Information

These products are considered articles under both U.S. and international product regulations and as such, these products do not require registration or notification on the various country-specific inventories.

B: Component Analysis - WHMIS IDL

This material has no components listed.

WHMIS Classification

This is not a WHMIS controlled product. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations. This SDS contains all the information required by the Controlled Products Regulations.

Section 16 - Other Information

Other Information

Prepared for:
Johns Manville
Insulation Systems
P. O. Box 5108
Denver, CO USA 80217-5108

Prepared by:
Johns Manville Technical Center
P.O. Box 625005
Littleton, CO USA 80162-5005

The information herein is presented in good faith and believed to be accurate as of the effective date given. However, no warranty, expressed or implied, is given. It is the buyer's responsibility to ensure that its activities comply with Federal, State or provincial, and local laws.

Date	MSDS #	Reason
08/01/00	2001-1.0000	New MSDS authoring system.
12/14/00	2001-1.0001	Minor editing.
07/16/03	2001-1.0002	Combined MSDS 2117 with this MSDS.
01/07/05	2001-1.0003	Addition of PVC Drain Pans to Sect. 1.
02/01/08	2001-1.0004	Regulatory update. Updated SDS to GHS format.
02/01/2011	2001-1.0005	Update MSDS.

End of Sheet 2001

MATERIAL SAFETY DATA SHEET

MSDS Number: 1101E

Section 1 **PRODUCT AND COMPANY IDENTIFICATION**

Trade Name: OATEY PVC MEDIUM CLEAR or GRAY CEMENT
Product Nos.: Clear -30350, 31017, 31018, 31019, 31020, 31021, 31550, 31551, 31552, 31553, 31946, 31947, 31948, 31949, 32222, 32223, 32224, 32225 Gray - 30883, 30884, 30885, 30886, 30887, 31930, 31931, 31932, 31933
Product Use: Cement for PVC Plastic Pipe
Formula: PVC Resin in Solvent Solution
Synonyms: PVC Plastic Pipe Cement
Firm Name & Address: Oatey Company 4700 West 160th Street, Cleveland, Ohio 44135 www.oatey.com
Firm Phone No: (216) 267-7100
Emergency Phone Nos.: For Emergency First Aid call 1-877-740-5015. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared by: Technical Department
Preparation Date: 10/01/2009

Section 2 **HAZARDS IDENTIFICATION**

Emergency Overview:
Clear or Gray
liquid with an ether-like odor. Extremely flammable liquid and vapor. Vapors may cause flash fire. May cause eye and skin irritation. Inhalation of vapors or mist may cause respiratory irritation and central nervous system effects. Swallowing may cause irritation, nausea, vomiting, diarrhea and kidney or liver disorders. Aspiration hazard. May be fatal if swallowed. Symptoms may be delayed.

Section 3 **COMPOSITION/INFORMATION ON INGREDIENTS**

INGREDIENTS:	%wt/wt :	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA	OTHER:
Tetrahydrofuran	30 – 50%	109-99-9	50 ppm(skin) 100 ppm STEL	200 ppm	25 ppm (Mfg)
Methyl Ethyl Ketone	10 – 25%	78-93-3	200 ppm 300 ppm	200 ppm	None
Acetone	10 – 25%	67-64-1	500 ppm 750 ppm STEL	1000 ppm	None
PVC Resin (Non-hazardous)	12 – 20%	9002-86-2	10 mg/m3	15 mg/m3	None
Cyclohexanone	10 – 20%	108-94-1	20 ppm(skin) 50 ppm STEL	50 ppm	None
Amorphous Fumed Silica (Non-hazardous)	1 - 5%	112945-52-5	10 mg/m3	None Established	None a

OSHA Hazard Classification: Flammable, irritant, organ effects

Section 4 **FIRST AID MEASURES**

Skin: Remove contaminated clothing immediately. Wash all exposed areas with soap and

	water. Get medical attention if irritation develops. Remove dried cement with hand cleaner or baby oil.
Eyes:	If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.
Inhalation:	If symptoms of exposure develop, remove to fresh air. If breathing becomes difficult, administer oxygen. Administer artificial respiration if breathing has stopped. Seek immediate medical attention.
Ingestion:	**DO NOT INDUCE VOMITING.** Rinse mouth with water. Never give anything by mouth to a person who is unconscious or drowsy. Get immediate medical attention by calling a Poison Control Center, or hospital emergency room. If medical advice cannot be obtained, then take the person and product to the nearest medical emergency treatment center or hospital.

Section 5	**FIRE FIGHTING MEASURES**
Flashpoint / Method:	14 - 23 Degrees F. (-10 to -5 Degrees C) / CCCFP
Flammability:	LEL = 1.8 % Volume, UEL = 11.8 % Volume
Extinguishing Media:	Use dry chemical, CO_2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure:	Firefighters should wear positive pressure self-contained breathing apparatus and full protective clothing for fires in areas where chemicals are used or stored
Unusual Fire And Explosion Hazards:	Extremely flammable liquid. Keep away from heat and all sources of ignition including sparks, flames, lighted cigarettes and pilot lights. Containers may rupture or explode in the heat of a fire. Vapors are heavier than air and may travel to a remote ignition source and flash back. This product contains tetrahydrofuran that may form explosive organic peroxide when exposed to air or light or with age.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.

Section 6	**ACCIDENTAL RELEASE MEASURES**
Spill or Leak Procedures:	Remove all sources of ignition and ventilate area. Stop leak if it can be done without risk. Personnel cleaning up the spill should wear appropriate personal protective equipment, including respirators if vapor concentrations are high. Soak up spill with an inert absorbent such as sand, earth or other non-combusting material. Put absorbent material in covered, labeled metal containers. Prevent liquid from entering watercourses, sewers and natural waterways. Report releases to authorities as required. See Section 13 for disposal information.

Section 7	**HANDLING AND STORAGE**
Handling:	Avoid contact with eyes, skin and clothing. Avoid breathing vapors or mists. Use with adequate ventilation (equivalent to outdoors). Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep product away from heat, sparks, flames and all other sources of ignition. No smoking in storage or use areas. Keep containers closed when not in use.
Storage:	Store in a cool, dry, well-ventilated area away from incompatible materials. Keep containers closed when not in use.
Other:	"Empty" containers retain product residue and can be hazardous. Follow all MSDS precautions in handling empty containers. Do not cut or weld on or near empty or full containers.

Section 8	**EXPOSURE CONTROLS/PERSONAL PROTECTION**
Ventilation:	Open doors & windows. Provide ventilation capable of maintaining emissions at the point of use below recommended exposure limits. If used in enclosed area, use exhaust fans. Exhaust fans should be explosion-proof or set up in a way

	that flammable concentrations of solvent vapors are not exposed to electrical fixtures or hot surfaces.
Respiratory Protection:	For operations where the exposure limit may be exceeded, a NIOSH approved organic vapor respirator or supplied air respirator is recommended. Equipment selection depends on contaminant type and concentration, select in accordance with 29 CFR 1910.134 and good industrial hygiene practice. For firefighting, use self-contained breathing apparatus.
Skin Protection:	Rubber gloves are suitable for normal use of the product. For long exposures chemical resistant gloves may be required such as 4H(tm) or Silver Shield(tm) to avoid prolonged skin contact.
Eye Protection:	Safety glasses with side shields or safety goggles.

Section 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	151 Degrees F / 66 Degrees C
Melting Point:	Not applicable
Vapor Pressure:	145 mmHg @ 20 Degrees C
Vapor Density:	(Air = 1) 2.5
Volatile Components:	80-84%
Solubility In Water:	Negligible
pH:	Not applicable
Specific Gravity:	0.93 +/- 0.02 @ 20 Degrees C
Evaporation Rate:	(BUAC = 1) = 5.5 - 8.0
Appearance:	Clear or Gray Liquid
Odor:	Ether-Like
Will Dissolve In:	Tetrahydrofuran
Material Is:	Liquid

Section 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, sparks, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.
Incompatibility/ Materials To Avoid:	Oxidizing agents, alkalis, amines, ammonia, acids, chlorine compounds, chlorinated inorganics (potassium, calcium and sodium hypochlorite) and hydrogen peroxides. May attack plastic, resins and rubber.
Hazardous Polymerization:	Will not occur.

Section 11 TOXICOLOGICAL INFORMATION

Inhalation:	Vapors or mists may cause mucous membrane and respiratory irritation, coughing, headache, dizziness, dullness, nausea, shortness of breath and vomiting. High concentrations may cause central nervous system depression, narcosis and unconsciousness. May cause kidney, liver and lung damage.
Skin:	May cause irritation with redness, itching and pain. Methyl ethyl ketone and cyclohexanone may be absorbed through the skin causing effects similar to those listed under inhalation.
Eye:	Vapors may cause irritation. Direct contact may cause irritation with redness, stinging and tearing of the eyes. May cause eye damage.
Ingestion:	Swallowing may cause abdominal pain, nausea, vomiting and diarrhea. Aspiration during swallowing or vomiting can cause chemical pneumonia and lung damage. May cause kidney and liver damage.
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis and damage to the kidney, liver, lungs and central nervous system.

Toxicity Data:	Acetone:	Oral rat LD50: 5,800 mg/kg Inhalation rat LC50: 50,100 mg/m3/8 hours
	Cyclohexanone:	Oral rat LD50: 1,620 mg/kg Inhalation rat LC50: 8,000 ppm/4 hours Skin rabbit LD50: 1 mL/kg
	Tetrahydrofuran:	Oral rat LD50: 1,650 mg/kg Inhalation rat LC50: 21,000 ppm/3 hours
	Methyl Ethyl Ketone:	Oral rat LD50: 2,737 mg/kg Inhalation rat LC50: 23,500 mg/m3/8 hours Skin rabbit LD50: 6,480 mg/kg

Sensitization: None of the components are known to cause sensitization.

Carcinogenicity: None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA. The National Toxicology Program has reported that exposure of mice and rats to tetrahydrofuran (THF) vapor levels up to 1800 ppm 6 hr/day, 5 days/week for their lifetime caused an increased incidence of kidney tumors in male rats and liver tumors in female mice. The significance of these findings for human health is unclear at this time, and may be related to "species specific" effects. Elevated incidences of tumors in humans have not been reported for THF. ACGIH has classified cyclohexanone (CYH) and tetrahydrofuran as “A3,” Confirmed Animal Carcinogens with Unknown Relevance to Humans.

Mutagenicity: Cyclohexanone has been positive in bacterial and mammalian assays. Acetone, methyl ethyl ketone and tetrahydrofuran are generally thought not to be mutagenic.

Reproductive Toxicity: Methyl ethyl ketone and cyclohexanone have been shown to cause embryofetal toxicity and birth defects in laboratory animals. Acetone and tetrahydrofuran has been found to cause adverse developmental effects only when exposure levels cause other toxic effects to the mother.

Medical Conditions Aggravated By Exposure: Persons with pre-existing skin, lung, kidney or liver disorders may be at increased risk from exposure to this product.

Section 12 ECOLOGICAL INFORMATION

This product is not expected to be toxic to aquatic organisms.
Cyclohexanone: 96 hour LC50 values for fish is over 100 mg/l.
Tetrahydrofuran: 96 hour LC50 fathead minnow: 2160 mg/L.
Acetone: 96 hour LC50 for fish is greater than 100 mg/L.
Methyl Ethyl Ketone: 96 hour LC50 for fish is greater than 100 mg/L.

VOC Information: This product emits VOC’s (volatile organic compounds) in its use. Make sure that use of this product complies with local VOC emission regulations, where they exist.

VOC Level: Maximum 510 g/L per SCAQMD Test Method 316A.

Section 13 DISPOSAL CONSIDERATIONS

Waste Disposal:	Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number:	U002, U057, U159, U213
EPA Hazardous Waste ID Number:	D001, D035, F003, F0005
EPA Hazard Waste Number:	Ignitable Waste. Toxic Waste (Methyl Ethyl Ketone content)

Section 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)

UN/NA Number:	None	UN1133
Proper Shipping Name:	Consumer Commodity	Adhesives
Hazard Class:	ORM-D	3
Packing Group:	None	PGII
Hazard Labels:	None	Flammable Liquid
IMDG		
UN Number:	UN1133	UN1133
Proper Shipping Name:	Adhesives	Adhesives
Hazard Class:	3	3
Packing Group:	II	II
Label:	None (Limited Quantities are expected from labeling)	Class 3 (Flammable Liquid)
Flashpoint (deg C)	-10 to -5 Degrees C	-10 to -5 Degrees C

2008 North American Emercency Response Guidebook Number: 127

Section 15 REGULATORY INFORMATION

Hazard Category for Section 311/312:	Acute Health, Chronic Health, Flammable
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain chemicals subject to SARA Title III Section 313 Reporting requirements.
CERCLA 103 Reportable Quantity:	Spills of this product over the RQ (reportable quantity) must be reported to the National Response Center. The RQ for the product, based on the RQ for Tetrahydrofuran (50% maximum) of 1,000 lbs, is 2,000 lbs. Many states have more stringent release reporting requirements. Report spills required under federal, state and local regulations.
California Proposition 65:	This product contains trace amounts of chemicals known to the State of California to cause cancer. Under normal use conditions, exposure to these chemicals at levels above the State of California “No Significant Risk Level” (NSRL) are unlikely. The use of proper personal protective equipment (PPE) and ventilation guidelines noted in Section 8 will minimize exposure to these chemicals.
TSCA Inventory	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Class B, Division 2; Class D, Division 2, Subdivision B; Class D, Division 2, Subdivision A. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

Section 16 OTHER INFORMATION

NFPA and HMIS:	
NFPA Hazard Signal:	Health: 2 Flammability: 3 Reactivity: 1 Special: None
HMIS Hazard Signal:	Health: 2* Flammability: 3 Reactivity: 1 PPE: G

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, we cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

Template: tmpl-us-e1

MATERIAL SAFETY DATA SHEET

SECTION 1 PRODUCT AND COMPANY IDENTIFICATION

Trade Name: OATEY PLUMBER'S PUTTY
Product No.: 31166, 31170, 31174
Product Use: Stainless fixture setting compound.
Formula: See SECTION 2
Synonyms: Putty.
Firm Name & Mailing Address: OATEY CO. 4700 West 160th Street P.O. Box 35906 Cleveland, Ohio 44135, U.S.A. http://www.oatey.com
Oatey Phone Number: (216) 267-7100 or (800) 321-9532
Emergency Phone Numbers: For Emergency First Aid call 1-303-623-5716 COLLECT. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the US: 1-703-527-3887.
Prepared By: Corporate Director – Safety and Environmental Compliance
Preparation Date: April 15, 2008

SECTION 2 COMPOSITION/INFORMATION ON INGREDIENTS

INGREDIENTS:	%wt/wt:	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA:	OTHER:
Limestone	60 - 90%	1317-65-3	10 mg/M3	15 mg/M3 (Total dust)	None
Hydrocarbon oil	10 - 30%	64742-52-5	5 mg/M3	5 mg/M3	None
Talc	1 - 5%	14807-96-6	2 mg/M3	2 mg/M3	None
Clay	1 – 5%	68953-58-2	10 mg/M3	10 mg/M3	None

OSHA Hazard Classification: Not hazardous

SECTION 3 HAZARDS IDENTIFICATION

Emergency Overview:
White putty with a mild petroleum odor. May cause mechanical irritation of the eyes.

SECTION 4 FIRST AID MEASURES

CALL 1-303-623-5716 COLLECT

Skin: Wash all exposed areas with soap and water.
Eyes: If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.
Inhalation: Not a likely route of entry.
Ingestion: **DO NOT INDUCE VOMITING.** Rinse mouth with water. Call a physician.

SECTION 5 FIRE FIGHTING MEASURES

Flashpoint / Method: Not applicable
Flammability: LEL = ND, UEL = ND
Extinguishing Media: Use dry chemical, CO_2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure: As appropriate for surrounding fire.
Unusual Fire and Explosion Hazards: None
Hazardous Decomposition Products: Combustion will produce toxic and irritating vapors including carbon monoxide and carbon dioxide.

SECTION 6 ACCIDENTAL RELEASE MEASURES

Spill or Leak Procedures: Sweep up dried material and dispose of with ordinary trash.

SECTION 7 HANDLING AND STORAGE

Handling: No special handling required. Wash thoroughly after handling. Use good hygiene practices.

Storage: Store in a cool, dry area.

Other: None.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Ventilation: Use with adequate ventilation.

Respiratory Protection: If putty dries and becomes dusty, wear a dust respirator during clean-up.

Skin Protection: Gloves are not normally required.

Eye Protection: Safety glasses with side shields are recommended.

Other: Eye wash should be available.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	Not applicable
Melting Point:	Not applicable
Vapor Pressure:	Not determined
Vapor Density:	Not determined
Volatile Components:	Not determined
Solubility In Water:	Negligible.
pH:	Not applicable
Specific Gravity:	2.0 @ 20 Degrees C
Evaporation Rate:	Not applicable
Appearance:	White Putty
Odor:	Petroleum odor
Will Dissolve In:	Not determined
Material Is:	Putty

SECTION 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide and carbon dioxide.
Incompatibility/ Materials To Avoid:	Strong oxidizing agents.
Hazardous Polymerization:	Will not occur.

SECTION 11 TOXICOLOGICAL INFORMATION

Inhalation:	Not a likely route of entry.
Skin:	May cause mild irritation. Prolonged and repeated contact may cause dermatitis.
Eye:	Contact may cause irritation.
Ingestion:	Swallowing may cause a laxative effect including cramps and diarrhea.
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis.
Toxicity Data:	Not available
Sensitization:	None of the components are known to cause sensitization.
Carcinogenicity:	None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA.
Mutagenicity:	None of the components are known to be mutagenic.
Reproductive Toxicity:	None of the components are known to be toxic to reproduction.
Medical Conditions Aggravated By Exposure:	None known.

SECTION 12 ECOLOGICAL INFORMATION

	No ecotoxicological information known.
VOC Information:	This product has very low VOC level.

SECTION 13 DISPOSAL CONSIDERATIONS

Waste Disposal: Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number: Not a hazardous waste.
EPA Hazardous Waste ID Number: Not a hazardous waste.
EPA Hazard Waste Class: Not a hazardous waste.

SECTION 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)
Proper Shipping Name:	Not regulated.	Not regulated.
Hazard Class/Packing Group:	None	None
UN/NA Number:	None	None
Hazard Labels:	None	None

IMDG	
Proper Shipping Name:	Not regulated
Hazard Class/Packing Group:	None
UN Number:	None
Label:	None

2004 North American Emergency Response Guidebook Number: None

SECTION 15 REGULATORY INFORMATION

Hazard Category for Section	Not hazardous.
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain any chemicals regulated under SARA Title III Section 313.
CERCLA 103 Reportable Quantity:	This product does not contain any chemicals regulated under CERCLA.
California Proposition 65:	This product does not contain any chemicals subject To California Proposition 65 regulation.
TSCA Inventory:	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Not a controlled product. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

SECTION 16 OTHER INFORMATION

NFPA and HMIS:

NFPA Hazard Signal:	Health: 0	Flammability: 0	Reactivity: 0	Special: None
HMIS Hazard Signal:	Health: 0	Flammability: 0	Reactivity: 0	PPE: A

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, Oatey cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

MATERIAL SAFETY DATA SHEET

MSDS Number: 1402E

Section 1 **PRODUCT AND COMPANY IDENTIFICATION**

Trade Name: OATEY PURPLE or CLEAR PRIMER NSF LISTED
Product Nos.: Purple - 30755, 30756, 30757, 30758, 30759, 30927 Clear - 30749, 30750, 30751, 30752, 30753, 30754, 31652, 31653
Product Use: Primer for PVC and CPVC Plastic Pipe
Formula: See section 2
Synonyms: Plastic Pipe Primer
Firm Name & Address: Oatey Company 4700 West 160th Street, Cleveland, Ohio 44135 www.oatey.com
Firm Phone No: (216) 267-7100
Emergency Phone Nos.: For Emergency First Aid call 1-877-740-5015. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared by: Technical Department
Preparation Date: 11/01/2009

Section 2 **HAZARDS IDENTIFICATION**

Emergency Overview:
Purple or Clear
liquid with an ether-like odor. Extremely flammable liquid and vapor. Vapors may cause flash fire. May cause eye and skin irritation. Inhalation of vapors or mist may cause respiratory irritation and central nervous system effects. Swallowing may cause irritation, nausea, vomiting, diarrhea and kidney or liver disorders. Aspiration hazard. May be fatal if swallowed. Symptoms may be delayed.

Section 3 **COMPOSITION/INFORMATION ON INGREDIENTS**

INGREDIENTS:	%wt/wt :	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA	OTHER:
Tetrahydrofuran	15 – 30%	109-99-9	50 ppm(skin) 100 ppm STEL	200 ppm	25 ppm (Mfg)
Methyl Ethyl Ketone	25 – 40%	78-93-3	200 ppm 300 ppm	200 ppm	None
Acetone	25 – 40%	67-64-1	500 ppm 750 ppm STEL	1000 ppm	None
Cyclohexanone	15 – 30%	108-94-1	20 ppm(skin) 50 ppm STEL	50 ppm	None

OSHA Hazard Classification: Flammable, irritant, organ effects

Section 4 **FIRST AID MEASURES**

Skin: Remove contaminated clothing immediately. Wash all exposed areas with soap and water. Get medical attention if irritation develops. Remove dried cement with hand cleaner or baby oil.
Eyes: If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.

Inhalation:	If symptoms of exposure develop, remove to fresh air. If breathing becomes difficult, administer oxygen. Administer artificial respiration if breathing has stopped. Seek immediate medical attention.
Ingestion:	**DO NOT INDUCE VOMITING.** Rinse mouth with water. Never give anything by mouth to a person who is unconscious or drowsy. Get immediate medical attention by calling a Poison Control Center, or hospital emergency room. If medical advice cannot be obtained, then take the person and product to the nearest medical emergency treatment center or hospital.

Section 5 FIRE FIGHTING MEASURES

Flashpoint / Method:	14 - 23 Degrees F. (-10 to -5 Degrees C) / CCCFP
Flammability:	LEL = 1.8 % Volume, UEL = 11.8 % Volume
Extinguishing Media:	Use dry chemical, CO_2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure:	Firefighters should wear positive pressure self-contained breathing apparatus and full protective clothing for fires in areas where chemicals are used or stored
Unusual Fire And Explosion Hazards:	Extremely flammable liquid. Keep away from heat and all sources of ignition including sparks, flames, lighted cigarettes and pilot lights. Containers may rupture or explode in the heat of a fire. Vapors are heavier than air and may travel to a remote ignition source and flash back. This product contains tetrahydrofuran that may form explosive organic peroxide when exposed to air or light or with age.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.

Section 6 ACCIDENTAL RELEASE MEASURES

Spill or Leak Procedures:	Remove all sources of ignition and ventilate area. Stop leak if it can be done without risk. Personnel cleaning up the spill should wear appropriate personal protective equipment, including respirators if vapor concentrations are high. Soak up spill with an inert absorbent such as sand, earth or other non-combusting material. Put absorbent material in covered, labeled metal containers. Prevent liquid from entering watercourses, sewers and natural waterways. Report releases to authorities as required. See Section 13 for disposal information.

Section 7 HANDLING AND STORAGE

Handling:	Avoid contact with eyes, skin and clothing. Avoid breathing vapors or mists. Use with adequate ventilation (equivalent to outdoors). Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep product away from heat, sparks, flames and all other sources of ignition. No smoking in storage or use areas. Keep containers closed when not in use.
Storage:	Store in a cool, dry, well-ventilated area away from incompatible materials. Keep containers closed when not in use.
Other:	"Empty" containers retain product residue and can be hazardous. Follow all MSDS precautions in handling empty containers. Do not cut or weld on or near empty or full containers.

Section 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Ventilation:	Open doors & windows. Provide ventilation capable of maintaining emissions at the point of use below recommended exposure limits. If used in enclosed area, use exhaust fans. Exhaust fans should be explosion-proof or set up in a way that flammable concentrations of solvent vapors are not exposed to electrical fixtures or hot surfaces.
Respiratory Protection:	For operations where the exposure limit may be exceeded, a NIOSH approved organic vapor respirator or supplied air respirator is recommended. Equipment selection depends on contaminant type and concentration, select in accordance

	with 29 CFR 1910.134 and good industrial hygiene practice. For firefighting, use self-contained breathing apparatus.
Skin Protection:	Rubber gloves are suitable for normal use of the product. For long exposures chemical resistant gloves may be required such as 4H(tm) or Silver Shield(tm) to avoid prolonged skin contact.
Eye Protection:	Safety glasses with side shields or safety goggles.

Section 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	151 Degrees F / 66 Degrees C
Melting Point:	Not applicable
Vapor Pressure:	145 mmHg @ 20 Degrees C
Vapor Density:	(Air = 1) 2.5
Volatile Components:	99.96%
Solubility In Water:	Negligible
pH:	Not applicable
Specific Gravity:	0.84 +/- 0.02 @ 20 Degrees C
Evaporation Rate:	(BUAC = 1) = 5.5 - 8.0
Appearance:	Purple or Clear Liquid
Odor:	Ether-Like
Will Dissolve In:	Tetrahydrofuran
Material Is:	Liquid

Section 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, sparks, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.
Incompatibility/ Materials To Avoid:	Oxidizing agents, alkalis, amines, ammonia, acids, chlorine compounds, chlorinated inorganics (potassium, calcium and sodium hypochlorite) and hydrogen peroxides. May attack plastic, resins and rubber.
Hazardous Polymerization:	Will not occur.

Section 11 TOXICOLOGICAL INFORMATION

Inhalation:	Vapors or mists may cause mucous membrane and respiratory irritation, coughing, headache, dizziness, dullness, nausea, shortness of breath and vomiting. High concentrations may cause central nervous system depression, narcosis and unconsciousness. May cause kidney, liver and lung damage.	
Skin:	May cause irritation with redness, itching and pain. Methyl ethyl ketone and cyclohexanone may be absorbed through the skin causing effects similar to those listed under inhalation.	
Eye:	Vapors may cause irritation. Direct contact may cause irritation with redness, stinging and tearing of the eyes. May cause eye damage.	
Ingestion:	Swallowing may cause abdominal pain, nausea, vomiting and diarrhea. Aspiration during swallowing or vomiting can cause chemical pneumonia and lung damage. May cause kidney and liver damage.	
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis and damage to the kidney, liver, lungs and central nervous system.	
Toxicity Data:	Acetone:	Oral rat LD50: 5,800 mg/kg Inhalation rat LC50: 50,100 mg/m3/8 hours
	Cyclohexanone:	Oral rat LD50: 1,620 mg/kg Inhalation rat LC50: 8,000 ppm/4 hours Skin rabbit LD50: 1 mL/kg

Tetrahydrofuran:	Oral rat LD50: 1,650 mg/kg Inhalation rat LC50: 21,000 ppm/3 hours
Methyl Ethyl Ketone:	Oral rat LD50: 2,737 mg/kg Inhalation rat LC50: 23,500 mg/m3/8 hours Skin rabbit LD50: 6,480 mg/kg

Sensitization: None of the components are known to cause sensitization.

Carcinogenicity: None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA. The National Toxicology Program has reported that exposure of mice and rats to tetrahydrofuran (THF) vapor levels up to 1800 ppm 6 hr/day, 5 days/week for their lifetime caused an increased incidence of kidney tumors in male rats and liver tumors in female mice. The significance of these findings for human health is unclear at this time, and may be related to "species specific" effects. Elevated incidences of tumors in humans have not been reported for THF. ACGIH has classified cyclohexanone (CYH) and tetrahydrofuran as “A3,” Confirmed Animal Carcinogens with Unknown Relevance to Humans.

Mutagenicity: Cyclohexanone has been positive in bacterial and mammalian assays. Acetone, methyl ethyl ketone and tetrahydrofuran are generally thought not to be mutagenic.

Reproductive Toxicity: Methyl ethyl ketone and cyclohexanone have been shown to cause embryofetal toxicity and birth defects in laboratory animals. Acetone and tetrahydrofuran has been found to cause adverse developmental effects only when exposure levels cause other toxic effects to the mother.

Medical Conditions Aggravated By Exposure: Persons with pre-existing skin, lung, kidney or liver disorders may be at increased risk from exposure to this product.

Section 12 ECOLOGICAL INFORMATION

This product is not expected to be toxic to aquatic organisms.
Cyclohexanone: 96 hour LC50 values for fish is over 100 mg/l.
Tetrahydrofuran: 96 hour LC50 fathead minnow: 2160 mg/L.
Acetone: 96 hour LC50 for fish is greater than 100 mg/L.
Methyl Ethyl Ketone: 96 hour LC50 for fish is greater than 100 mg/L.

VOC Information: This product emits VOC’s (volatile organic compounds) in its use. Make sure that use of this product complies with local VOC emission regulations, where they exist.

VOC Level: Maximum 550 g/L per SCAQMD Test Method 316A.

Section 13 DISPOSAL CONSIDERATIONS

Waste Disposal:	Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number:	U002, U057, U159, U213
EPA Hazardous Waste ID Number:	D001, D035, F003, F0005
EPA Hazard Waste Number:	Ignitable Waste. Toxic Waste (Methyl Ethyl Ketone content)

Section 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)
UN/NA Number:	None	UN1993
Proper Shipping Name:	Consumer Commodity	Flammable Liquid, NOS (Methyl Ethyl Ketone, Acetone)
Hazard Class:	ORM-D	3

Packing Group:	None	PGII
Hazard Labels:	None	Flammable Liquid
IMDG		
UN Number:	UN1993	UN1993
Proper Shipping Name:	Flammable Liquid, NOS (Limited Quantity)	Flammable Liquid, NOS (Methyl Ethyl Ketone, Acetone)
Hazard Class:	3	3
Packing Group:	II	II
Label:	None (Limited Quantities are expected from labeling)	Class 3 (Flammable Liquid)
Flashpoint (deg C)	-10 to -5 Degrees C	-10 to -5 Degrees C

2008 North American Emercency Response Guidebook Number: 127

Section 15 REGULATORY INFORMATION

Hazard Category for Section 311/312:	Acute Health, Chronic Health, Flammable
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain chemicals subject to SARA Title III Section 313 Reporting requirements.
CERCLA 103 Reportable Quantity:	Spills of this product over the RQ (reportable quantity) must be reported to the National Response Center. The RQ for the product, based on the RQ for Tetrahydrofuran (30% maximum) of 1,000 lbs, is 3,333 lbs. Many states have more stringent release reporting requirements. Report spills required under federal, state and local regulations.
California Proposition 65:	This product does not contain any chemicals subject to California Proposition 65 regulations.
TSCA Inventory	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Class B, Division 2; Class D, Division 2, Subdivision B; Class D, Division 2, Subdivision A. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

Section 16 OTHER INFORMATION

NFPA and HMIS:	
NFPA Hazard Signal:	Health: 2 Flammability: 3 Reactivity: 1 Special: None
HMIS Hazard Signal:	Health: 2* Flammability: 3 Reactivity: 1 PPE: G

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, we cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

Template: tmpl-us-e1

MATERIAL SAFETY DATA SHEET

SECTION 1 PRODUCT AND COMPANY IDENTIFICATION

Trade Name: OATEY GREAT WHITE PIPE JOINT COMPOUND WITH PTFE
Product No.: 31229, 31230, 31231, 31232, 31233, 31234, 31437, 48007, 48008, 48009, 48321, 48322
Product Use: Compound for sealing and lubricating threaded pipe joints.
Formula: See SECTION 2
Synonyms: Pipe dope.
Firm Name & Mailing Address: OATEY CO. 4700 West 160th Street P.O. Box 35906 Cleveland, Ohio 44135, U.S.A. http://www.oatey.com
Oatey Phone Number: (216) 267-7100 or (800) 321-9532
Emergency Phone Numbers: For Emergency First Aid call 1-303-623-5716 COLLECT. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared By: Corporate Director – Safety and Environmental Compliance
Preparation Date: April 29, 2008

SECTION 2 COMPOSITION/INFORMATION ON INGREDIENTS

INGREDIENTS:	%wt/wt:	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA:	OTHER:
Calcium Carbonate	35 - 45%	1317-65-3	10 mg/M3	15 mg/M3 (Total dust)	None
Hydrocarbon oil	30 - 40%	64742-52-5	5 mg/M3	5 mg/M3	None
Titanium Dioxide	5 – 10%	13463-67-7	10 mg/M3	15 mg/M3	None
Clay	1 – 5%	68953-58-2	10 mg/M3	10 mg/M3	None

OSHA Hazard Classification: Not hazardous

SECTION 3 HAZARDS IDENTIFICATION

Emergency Overview:
White putty with a mild petroleum odor. May cause mechanical irritation of the eyes.

SECTION 4 FIRST AID MEASURES

CALL 1-303-623-5716 COLLECT

Skin: Wash all exposed areas with soap and water.
Eyes: If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.
Inhalation: Not a likely route of entry.
Ingestion: **DO NOT INDUCE VOMITING.** Rinse mouth with water. Call a physician.

SECTION 5 FIRE FIGHTING MEASURES

Flashpoint / Method: Not applicable
Flammability: LEL = Not determined, UEL = Not determined
Extinguishing Media: Use dry chemical, CO_2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure: As appropriate for surrounding fire.
Unusual Fire and Explosion Hazards: None
Hazardous Decomposition Products: Combustion will produce toxic and irritating vapors including carbon monoxide and carbon dioxide.

SECTION 6 ACCIDENTAL RELEASE MEASURES

Spill or Leak Procedures: Sweep up dried material and dispose of with ordinary trash.

SECTION 7 HANDLING AND STORAGE

Handling: No special handling required. Wash thoroughly after handling.
Storage: Store in a cool, dry area.
Other: None.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Ventilation: Use with adequate ventilation. If putty dries and becomes dusty,
Respiratory Protection: wear a dust respirator during clean-up.
Skin Protection: Gloves are not normally required.
Eye Protection: Safety glasses with side shields are recommended.
Other: Eye wash should be available.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	Not applicable
Melting Point:	Not applicable
Vapor Pressure:	Not determined
Vapor Density:	Not determined
Volatile Components:	Not determined
Solubility In Water:	Negligible.
pH:	Not applicable
Specific Gravity:	2.0 @ 20 Degrees C
Evaporation Rate:	Not determined
Appearance:	White Paste
Odor:	Petroleum odor
Will Dissolve In:	Not determined
Material Is:	Paste

SECTION 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide and carbon dioxide.
Incompatibility/ Materials To Avoid:	Oxidizing agents.
Hazardous Polymerization:	Will not occur.

SECTION 11 TOXICOLOGICAL INFORMATION

Inhalation:	Not a likely route of entry.
Skin:	May cause mild irritation. Prolonged and repeated contact may cause dermatitis.
Eye:	Contact may cause irritation.
Ingestion:	Swallowing may cause a laxative effect including cramps and diarrhea.
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis. to the kidney, liver, lungs and central nervous system.
Toxicity Data:	Not available
Sensitization:	None of the components are known to cause sensitization.
Carcinogenicity:	None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA.
Mutagenicity:	None of the components are known to be mutagenic.
Reproductive Toxicity:	None of the components are known to be toxic to reproduction.
Medical Conditions Aggravated By Exposure:	None known.

SECTION 12 ECOLOGICAL INFORMATION

	No ecotoxicological information known.
VOC Information:	This product has very low VOC level.

SECTION 13 DISPOSAL CONSIDERATIONS

Waste Disposal: Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number: Not a hazardous waste.
EPA Hazardous Waste ID Number: Not a hazardous waste.
EPA Hazard Waste Class: Not a hazardous waste.

SECTION 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)
Proper Shipping Name:	Not regulated.	Not regulated.
Hazard Class/Packing Group:	None	None
UN/NA Number:	None	None
Hazard Labels:	None	None

IMDG	
Proper Shipping Name:	Not regulated
Hazard Class/Packing Group:	None
UN Number:	None
Label:	None

2004 North American Emergency Response Guidebook Number: None

SECTION 15 REGULATORY INFORMATION

Hazard Category for Section	Not hazardous.
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain any chemicals regulated under SARA Title III Section 313.
CERCLA 103 Reportable Quantity:	This product does not contain any chemicals regulated under CERCLA.
California Proposition 65:	This product does not contain any chemicals subject To California Proposition 65 regulation.
TSCA Inventory:	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Not a controlled product. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

SECTION 16 OTHER INFORMATION

NFPA and HMIS:

NFPA Hazard Signal:	Health: 0	Flammability: 0	Reactivity: 0	Special: None
HMIS Hazard Signal:	Health: 0	Flammability: 0	Reactivity: 0	PPE: A

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, Oatey cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

MATERIAL SAFETY DATA SHEET

MSDS Number: 1102E

Section 1 **PRODUCT AND COMPANY IDENTIFICATION**

Trade Name: OATEY PVC HEAVY DUTY CLEAR or GRAY CEMENT - LO-VOC FORMULA
Product Nos.: Clear 30850, 30863, 30876(TV), 30882, 31008(TV), 31011, 31950, 31951, 31952, 31953 Gray 30349, 31093, 31094, 31095, 31105, 31118, 31978, 31979, 31980, 31981, 32050, 32051, 32052, 32210, 32211
Product Use: Cement for PVC Plastic Pipe
Formula: PVC Resin in Solvent Solution
Synonyms: PVC Plastic Pipe Cement
Firm Name & Address: Oatey Company 4700 West 160th Street, Cleveland, Ohio 44135 www.oatey.com
Firm Phone No: (216) 267-7100
Emergency Phone Nos.: For Emergency First Aid call 1-877-740-5015. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared by: Technical Department
Preparation Date: 02/08/2010

Section 2 **HAZARDS IDENTIFICATION**

Emergency Overview:
Clear or Gray
liquid with an ether-like odor. Extremely flammable liquid and vapor. Vapors may cause flash fire. May cause eye and skin irritation. Inhalation of vapors or mist may cause respiratory irritation and central nervous system effects. Swallowing may cause irritation, nausea, vomiting, diarrhea and kidney or liver disorders. Aspiration hazard. May be fatal if swallowed. Symptoms may be delayed.

Section 3 **COMPOSITION/INFORMATION ON INGREDIENTS**

INGREDIENTS:	%wt/wt :	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA	OTHER:
Tetrahydrofuran	40 - 60%	109-99-9	50 ppm(skin) 100 ppm STEL	200 ppm	25 ppm (Mfg)
Methyl Ethyl Ketone	5 – 15%	78-93-3	200 ppm 300 ppm	200 ppm	None
Acetone	10 – 25%	67-64-1	500 ppm 750 ppm STEL	1000 ppm	None
PVC Resin (Non-hazardous)	12 – 20%	9002-86-2	10 mg/m3	15 mg/m3	None
Cyclohexanone	10 – 25%	108-94-1	20 ppm(skin) 50 ppm STEL	50 ppm	None
Amorphous Fumed Silica (Non-hazardous)	1 - 4%	112945-52-5	10 mg/m3	None Established	None a

OSHA Hazard Classification: Flammable, irritant, organ effects

Section 4 **FIRST AID MEASURES**

Skin: Remove contaminated clothing immediately. Wash all exposed areas with soap and

	water. Get medical attention if irritation develops. Remove dried cement with hand cleaner or baby oil.
Eyes:	If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.
Inhalation:	If symptoms of exposure develop, remove to fresh air. If breathing becomes difficult, administer oxygen. Administer artificial respiration if breathing has stopped. Seek immediate medical attention.
Ingestion:	**DO NOT INDUCE VOMITING.** Rinse mouth with water. Never give anything by mouth to a person who is unconscious or drowsy. Get immediate medical attention by calling a Poison Control Center, or hospital emergency room. If medical advice cannot be obtained, then take the person and product to the nearest medical emergency treatment center or hospital.

Section 5	**FIRE FIGHTING MEASURES**
Flashpoint / Method:	14 - 23 Degrees F. (-10 to -5 Degrees C) / CCCFP
Flammability:	LEL = 1.8 % Volume, UEL = 11.8 % Volume
Extinguishing Media:	Use dry chemical, CO_2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure:	Firefighters should wear positive pressure self-contained breathing apparatus and full protective clothing for fires in areas where chemicals are used or stored
Unusual Fire And Explosion Hazards:	Extremely flammable liquid. Keep away from heat and all sources of ignition including sparks, flames, lighted cigarettes and pilot lights. Containers may rupture or explode in the heat of a fire. Vapors are heavier than air and may travel to a remote ignition source and flash back. This product contains tetrahydrofuran that may form explosive organic peroxide when exposed to air or light or with age.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.

Section 6	**ACCIDENTAL RELEASE MEASURES**
Spill or Leak Procedures:	Remove all sources of ignition and ventilate area. Stop leak if it can be done without risk. Personnel cleaning up the spill should wear appropriate personal protective equipment, including respirators if vapor concentrations are high. Soak up spill with an inert absorbent such as sand, earth or other non-combusting material. Put absorbent material in covered, labeled metal containers. Prevent liquid from entering watercourses, sewers and natural waterways. Report releases to authorities as required. See Section 13 for disposal information.

Section 7	**HANDLING AND STORAGE**
Handling:	Avoid contact with eyes, skin and clothing. Avoid breathing vapors or mists. Use with adequate ventilation (equivalent to outdoors). Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep product away from heat, sparks, flames and all other sources of ignition. No smoking in storage or use areas. Keep containers closed when not in use.
Storage:	Store in a cool, dry, well-ventilated area away from incompatible materials. Keep containers closed when not in use.
Other:	"Empty" containers retain product residue and can be hazardous. Follow all MSDS precautions in handling empty containers. Do not cut or weld on or near empty or full containers.

Section 8	**EXPOSURE CONTROLS/PERSONAL PROTECTION**
Ventilation:	Open doors & windows. Provide ventilation capable of maintaining emissions at the point of use below recommended exposure limits. If used in enclosed area, use exhaust fans. Exhaust fans should be explosion-proof or set up in a way

	that flammable concentrations of solvent vapors are not exposed to electrical fixtures or hot surfaces.
Respiratory Protection:	For operations where the exposure limit may be exceeded, a NIOSH approved organic vapor respirator or supplied air respirator is recommended. Equipment selection depends on contaminant type and concentration, select in accordance with 29 CFR 1910.134 and good industrial hygiene practice. For firefighting, use self-contained breathing apparatus.
Skin Protection:	Rubber gloves are suitable for normal use of the product. For long exposures chemical resistant gloves may be required such as 4H(tm) or Silver Shield(tm) to avoid prolonged skin contact.
Eye Protection:	Safety glasses with side shields or safety goggles.

Section 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	151 Degrees F / 66 Degrees C
Melting Point:	Not applicable
Vapor Pressure:	145 mmHg @ 20 Degrees C
Vapor Density:	(Air = 1) 2.5
Volatile Components:	80-84%
Solubility In Water:	Negligible
pH:	Not applicable
Specific Gravity:	0.94 +/- 0.02 @ 20 Degrees C
Evaporation Rate:	(BUAC = 1) = 5.5 - 8.0
Appearance:	Clear or Gray Liquid
Odor:	Ether-Like
Will Dissolve In:	Tetrahydrofuran
Material Is:	Liquid

Section 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, sparks, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.
Incompatibility/ Materials To Avoid:	Oxidizing agents, alkalis, amines, ammonia, acids, chlorine compounds, chlorinated inorganics (potassium, calcium and sodium hypochlorite) and hydrogen peroxides. May attack plastic, resins and rubber.
Hazardous Polymerization:	Will not occur.

Section 11 TOXICOLOGICAL INFORMATION

Inhalation:	Vapors or mists may cause mucous membrane and respiratory irritation, coughing, headache, dizziness, dullness, nausea, shortness of breath and vomiting. High concentrations may cause central nervous system depression, narcosis and unconsciousness. May cause kidney, liver and lung damage.
Skin:	May cause irritation with redness, itching and pain. Methyl ethyl ketone and cyclohexanone may be absorbed through the skin causing effects similar to those listed under inhalation.
Eye:	Vapors may cause irritation. Direct contact may cause irritation with redness, stinging and tearing of the eyes. May cause eye damage.
Ingestion:	Swallowing may cause abdominal pain, nausea, vomiting and diarrhea. Aspiration during swallowing or vomiting can cause chemical pneumonia and lung damage. May cause kidney and liver damage.
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis and damage to the kidney, liver, lungs and central nervous system.

Toxicity Data:	Acetone:	Oral rat LD50: 5,800 mg/kg Inhalation rat LC50: 50,100 mg/m3/8 hours
	Cyclohexanone:	Oral rat LD50: 1,620 mg/kg Inhalation rat LC50: 8,000 ppm/4 hours Skin rabbit LD50: 1 mL/kg
	Tetrahydrofuran:	Oral rat LD50: 1,650 mg/kg Inhalation rat LC50: 21,000 ppm/3 hours
	Methyl Ethyl Ketone:	Oral rat LD50: 2,737 mg/kg Inhalation rat LC50: 23,500 mg/m3/8 hours Skin rabbit LD50: 6,480 mg/kg

Sensitization: None of the components are known to cause sensitization.

Carcinogenicity: None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA. The National Toxicology Program has reported that exposure of mice and rats to tetrahydrofuran (THF) vapor levels up to 1800 ppm 6 hr/day, 5 days/week for their lifetime caused an increased incidence of kidney tumors in male rats and liver tumors in female mice. The significance of these findings for human health is unclear at this time, and may be related to "species specific" effects. Elevated incidences of tumors in humans have not been reported for THF. ACGIH has classified cyclohexanone (CYH) and tetrahydrofuran as “A3,” Confirmed Animal Carcinogens with Unknown Relevance to Humans.

Mutagenicity: Cyclohexanone has been positive in bacterial and mammalian assays. Acetone, methyl ethyl ketone and tetrahydrofuran are generally thought not to be mutagenic.

Reproductive Toxicity: Methyl ethyl ketone and cyclohexanone have been shown to cause embryofetal toxicity and birth defects in laboratory animals. Acetone and tetrahydrofuran has been found to cause adverse developmental effects only when exposure levels cause other toxic effects to the mother.

Medical Conditions Aggravated By Exposure: Persons with pre-existing skin, lung, kidney or liver disorders may be at increased risk from exposure to this product.

Section 12 ECOLOGICAL INFORMATION

This product is not expected to be toxic to aquatic organisms.
Cyclohexanone: 96 hour LC50 values for fish is over 100 mg/l.
Tetrahydrofuran: 96 hour LC50 fathead minnow: 2160 mg/L.
Acetone: 96 hour LC50 for fish is greater than 100 mg/L.
Methyl Ethyl Ketone: 96 hour LC50 for fish is greater than 100 mg/L.

VOC Information: This product emits VOC’s (volatile organic compounds) in its use. Make sure that use of this product complies with local VOC emission regulations, where they exist.

VOC Level: Maximum 510 g/L per SCAQMD Test Method 316A.

Section 13 DISPOSAL CONSIDERATIONS

Waste Disposal:	Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number:	U002, U057, U159, U213
EPA Hazardous Waste ID Number:	D001, D035, F003, F0005
EPA Hazard Waste Number:	Ignitable Waste. Toxic Waste (Methyl Ethyl Ketone content)

Section 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)

UN/NA Number:	None	UN1133
Proper Shipping Name:	Consumer Commodity	Adhesives
Hazard Class:	ORM-D	3
Packing Group:	None	PGII
Hazard Labels:	None	Flammable Liquid
IMDG		
UN Number:	UN1133	UN1133
Proper Shipping Name:	Adhesives	Adhesives
Hazard Class:	3	3
Packing Group:	II	II
Label:	None (Limited Quantities are expected from labeling)	Class 3 (Flammable Liquid)
Flashpoint (deg C)	-10 to -5 Degrees C	-10 to -5 Degrees C

2008 North American Emercency Response Guidebook Number: 127

Section 15 REGULATORY INFORMATION

Hazard Category for Section 311/312:	Acute Health, Chronic Health, Flammable
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain chemicals subject to SARA Title III Section 313 Reporting requirements.
CERCLA 103 Reportable Quantity:	Spills of this product over the RQ (reportable quantity) must be reported to the National Response Center. The RQ for the product, based on the RQ for Tetrahydrofuran (60% maximum) of 1,000 lbs, is 1,667 lbs. Many states have more stringent release reporting requirements. Report spills required under federal, state and local regulations.
California Proposition 65:	This product contains trace amounts of chemicals known to the State of California to cause cancer. Under normal use conditions, exposure to these chemicals at levels above the State of California “No Significant Risk Level” (NSRL) are unlikely. The use of proper personal protective equipment (PPE) and ventilation guidelines noted in Section 8 will minimize exposure to these chemicals.
TSCA Inventory	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Class B, Division 2; Class D, Division 2, Subdivision B; Class D, Division 2, Subdivision A. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

Section 16 OTHER INFORMATION

NFPA and HMIS:	
NFPA Hazard Signal:	Health: 2 Flammability: 3 Reactivity: 1 Special: None
HMIS Hazard Signal:	Health: 2* Flammability: 3 Reactivity: 1 PPE: G

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, we cannot give any guarantees regarding information from other sources, and expressly do not make warranties, nor assume any liability for its use.

Template: tmpl-us-e1

Material Safety Data Sheet

Section 1 Product and Company Identification

Product Name: **FLUX-RITE 90 PASTE**

Revision #: 1.7 **Date Prepared:** December 2, 1994 **Date Revised:** February 4, 2010

Manufacturer:

LA-CO INDUSTRIES, Inc./Markal Co.
1201 Pratt Blvd.
Elk Grove Village, IL, USA 60007-5746
Information Telephone: 847-956-7600
Emergency Telephone: Call CHEMTREC
USA 800-424-9300
International (Call Collect) 1-703-527-3887
Chemical Formula: Mixture
CAS No.: Not Applicable. **Derivation:** Not Applicable.
Synonyms: Not Applicable.
General Use: ASTM B 813 Compliant Soldering Flux for Copper, Brass, Tin, and Steel.

Supplier/Importer:

Section 2 Composition/Information on Ingredients

Ingredient	CAS No.	%
Ammonium Chloride [3,4,5,6]	12125-02-9	2

For Fume - ACGIH: TWA = 10mg\m^3 STEL = 20mg\m^3
EPA: CERCLA RQ = 5000 lbs.

(For Section 2 footnotes: See Section 15)

Section 3 Hazards Identification

EMERGENCY OVERVIEW: Caution: Soldering may produce irritating fumes and gases. Use adequate ventilation.

POTENTIAL HEALTH EFFECTS:
Primary Entry Routes: Ingestion, Inhalation
Acute Effects
Eyes: Fumes evolved during soldering may be irritating.
Skin: Not Applicable.
Ingestion: Not Applicable.
Inhalation: Fumes evolved during soldering may be irritating.
Chronic Effects
Eyes: Not Applicable.
Skin: Not Applicable.
Ingestion: Not Applicable.
Inhalation: Not Applicable.
Carcinogenicity: Not Applicable.
Target Organ Effects: Not Applicable.
Medical Conditions Aggravated by Long-Term Exposure: Not Determined.
Other Information: BASED ON RESULTS OF INDEPENDENT TESTS AND IN ACCORDANCE WITH THE U.S. FEDERAL HAZARDOUS SUBSTANCES LABELING ACT AND U.S. CONSUMER PRODUCT SAFETY COMMISSION REGULATIONS, THIS PRODUCT IS NOT AN EYE OR SKIN IRRITANT, NOR IS IT TOXIC BY ORAL INGESTION.
HMIS Rating: Health 0, Flammability 1, Reactivity 0

Section 4 First Aid

Eye Contact: Flush with water.
Skin Contact: Rinse with water.
Ingestion: Drink water if irritation develops.
Inhalation: Remove from exposure.
Other Information: Consult physician if any exposure symptoms persist.

Section 5 Fire Fighting Measures

Flash Point (method): Not Applicable.
Autoignition Temperature: Not Applicable.
LEL: Not Applicable. **UEL:** Not Applicable.
Flammability Classification: Not Applicable.
Extinguishing Media: Water fog, alcohol foam, dry chemical.
Hazardous Combustion Products: Carbon Monoxide, Carbon Dioxide, Ammonia, Ammonium Chloride, Hydrogen Chloride.
Unusual Fire or Explosion Hazards: Not Applicable.
Fire-Fighting Instructions/Equipment: Keep personnel removed and upwind of any fire. Wear full fire-fighting turn-out gear (full Bunker gear), and respiratory protection (SCBA).
NFPA Rating: Health 1, Flammability 1, Reactivity 0

Section 6 Accidental Release Measures

Use recommended personal protective equipment (see Section 8).
Small Spill: Scrape or wipe up. Flush remainder with water.
Large Spill: Contain spill. Scrape up for reuse or soak up with absorbent material. Flush remainder with water.

Section 7 Handling and Storage

Handling Precautions: Use recommended personal protective equipment (see Section 8). Wash thoroughly after handling.
Storage Requirements: Store in a cool, dry area.

Section 8 Exposure Controls/Personal Protection

Eye/Face Protection: Goggles or face shield recommended to protect from burns when soldering.
Skin Protection: Gloves recommended to protect from burns when soldering.
Respiratory Protection: Acid gas respirator in very enclosed areas.
Other Personal Protective Equipment: Eye wash and safety shower.
Engineering Controls: Normal room ventilation. Local exhaust in confined areas.
Administrative Controls: Users of this product must be properly trained and qualified in its use.
Other Information: Not Applicable.

Section 9 Physical and Chemical Properties

Appearance/Physical State: LIQUID: Green liquid. PASTE: Pale yellow gel.

Odor: Low odor.
Odor Threshold (ppm): Not Determined.
Specific Gravity (H_2O = 1): Paste: 1.1, Liquid: 1.18 (9.80 lbs./gal.)
Solubility in Water: Soluble.
Coefficient of Water/Oil Solubility: >>1
pH: 7
Melting Point: Not Applicable.
Boiling Point: Not Applicable.
Vapor Pressure (mm Hg at 20°C): Not Applicable.
Vapor Density (Air = 1): Not Applicable.
Evaporation Rate (n-BuAc=1): <<1
V.O.C.: 0%(w/w), 0%(v/v), 0 lbs./gal., 0 kg/l

Section 10 Stability and Reactivity

Chemical Stability: Stable.
Hazardous Polymerization: Will not occur.
Conditions to Avoid: Not Applicable.
Chemicals to Avoid: Oxidizers, Cyanides, Sulfides.
Hazardous Decomposition Products: Not Applicable.

Section 11 Toxicological Information

Sensitization to Product: Not Applicable.
Irritancy of Product: Not Applicable.
Reproductive Toxicity: Not Applicable.
Teratogenicity: Not Applicable.
Mutagenicity: Not Applicable.

Toxicological information regarding individual ingredients, if applicable, may be found in Section 2.

Section 12 Ecological Information

Not Determined.

Section 13 Disposal Considerations

Dispose of in accordance with applicable federal, state, and local regulations.

Section 14 Transport Information

D.O.T. (U.S.)
Proper Shipping Name: Not Regulated.
Hazard Class or Division: Not Regulated.
Hazard Label: Not Regulated.
I.D. Number: Not Regulated.
TDG (Canada): Not Regulated.
IATA: Not Regulated.
ICAO: Not Regulated.
IMO: Not Regulated.

Australian Code for the Transport of Dangerous Goods
Dangerous Goods Class and Subsidiary Risk: Not Determined.

Section 15 Regulatory Information

Footnotes for Section 2:
1 Subject to the reporting requirements of SARA Title III, Section 313.
2 Appears on the California Safe Drinking Water and Toxic Enforcement Act (Prop. 65) Substances List.
3 Appears on the Massachusetts Substances List.
4 Appears on the New Jersey Right-To-Know Hazardous Substances List.
5 Appears on the Pennsylvania Hazardous Substances List.
6 Appears on the Canadian WHMIS Ingredient Disclosure List.

U.S.A.

OSHA Hazard Status: This product is not considered to be hazardous as defined by the U.S. OSHA HCS (29 CFR 1910.1200).
EPA SARA sec. 311/312 Hazard Categories: Not Applicable.
Toxic Substances Control Act (TSCA): All ingredients contained in this product are listed on the U.S. EPA TSCA Chemical Substance Inventory.
HMIS Rating: Health 0, Flammability 1, Reactivity 0
NFPA Rating: Health 1, Flammability 1, Reactivity 0

American Society for Testing and Materials (ASTM): FLUX-RITE 90 meets ASTM Spec. B813.
Copper Development Association (CDA): FLUX-RITE 90 meets CDA Test Method 1.0.

CANADA

WHMIS Status: This product is not considered to be hazardous as defined by Canadian WHMIS Controlled Products Regulations.
WHMIS Rating: None.
WHMIS Risk Phrases: None.
WHMIS Precautionary Statements: None.
Domestic Substances List (DSL): All ingredients contained in this product are listed on the Canadian EPA (CEPA) Domestic Substances List (DSL).

E.U.

European Inventory of Existing Chemical Substances (EINECS): All ingredients contained in this product are listed on the European Inventory of Existing Chemical Substances (EINECS).
Categories of Danger (Labeling Information): None.
Risk (R) Phrases: None.
Safety (S) Phrases: Keep out of reach of children (S2)

AUSTRALIA

Worksafe Australia Status: This product is not classified as hazardous according to criteria of Worksafe Australia.
HAZCHEM Code: None allocated.
Poisons Schedule Number: None allocated.

Further regulatory information regarding individual ingredients, if applicable, may be found in Section 2.

This product has been classified in accordance with the hazard criteria of the U.S. OSHA Hazard Communication Standard, the Canadian WHMIS Controlled Products Regulations, the British CHIP2 regulation 6, and the Australian NMRCWHS. This MSDS contains the information required by the above regulations and conforms to ANSI Z400.1-1993.

Section 16 Other Information

MSDS Prepared By: Director of Chemical Safety

The information contained herein is based on data available to us and is accurate and reliable to the best of our knowledge and belief. However, LA-CO Industries, Inc. makes no representations as to its completeness or accuracy. Information is supplied on condition that persons receiving such information will make their own determination as to its suitability for their purposes prior to use. In no event will LA-CO Industries, Inc. be responsible for damages of any nature whatsoever resulting from the use of or reliance upon the information contained herein.

MATERIAL SAFETY DATA SHEET

MSDS Number: 1203E

Section 1 **PRODUCT AND COMPANY IDENTIFICATION**

Trade Name: OATEY CPVC FLOWGUARD GOLD ONE-STEP YELLOW CEMENT
Product Nos.: 31910, 31911, 31912, 31913, 31914, 31656, 31657, 32200, 32201, 32202, 32203, 31660, 31661, 31662, 31663 UVI - 31917, 31918, 31919
Product Use: Cement for CPVC Plastic Pipe
Formula: CPVC Resin in Solvent Solution
Synonyms: CPVC Plastic Pipe Cement
Firm Name & Address: Oatey Company 4700 West 160th Street, Cleveland, Ohio 44135 www.oatey.com
Firm Phone No: (216) 267-7100
Emergency Phone Nos.: For Emergency First Aid call 1-877-740-5015. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared by: Technical Department
Preparation Date: 11/01/2009

Section 2 **HAZARDS IDENTIFICATION**

Emergency Overview:
Yellow/Gold
liquid with an ether-like odor. Extremely flammable liquid and vapor. Vapors may cause flash fire. May cause eye and skin irritation. Inhalation of vapors or mist may cause respiratory irritation and central nervous system effects. Swallowing may cause irritation, nausea, vomiting, diarrhea and kidney or liver disorders. Aspiration hazard. May be fatal if swallowed. Symptoms may be delayed.

Section 3 **COMPOSITION/INFORMATION ON INGREDIENTS**

INGREDIENTS:	%wt/wt :	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA	OTHER:
Tetrahydrofuran	30 – 50%	109-99-9	50 ppm(skin) 100 ppm STEL	200 ppm	25 ppm (Mfg)
Methyl Ethyl Ketone	10 – 30%	78-93-3	200 ppm 300 ppm	200 ppm	None
Acetone	5 – 20%	67-64-1	500 ppm 750 ppm STEL	1000 ppm	None
CPVC Resin (Non-hazardous)	12 – 20%	68648-82-8	10 mg/m3	None Established	None
Cyclohexanone	5 – 15%	108-94-1	20 ppm(skin) 50 ppm STEL	50 ppm	None
Amorphous Fumed Silica (Non-hazardous)	1 - 5%	112945-52-5	10 mg/m3	None Established	None a

OSHA Hazard Classification: Flammable, irritant, organ effects

Section 4 **FIRST AID MEASURES**

Skin: Remove contaminated clothing immediately. Wash all exposed areas with soap and water. Get medical attention if irritation develops. Remove dried cement with

	hand cleaner or baby oil.
Eyes:	If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.
Inhalation:	If symptoms of exposure develop, remove to fresh air. If breathing becomes difficult, administer oxygen. Administer artificial respiration if breathing has stopped. Seek immediate medical attention.
Ingestion:	**DO NOT INDUCE VOMITING.** Rinse mouth with water. Never give anything by mouth to a person who is unconscious or drowsy. Get immediate medical attention by calling a Poison Control Center, or hospital emergency room. If medical advice cannot be obtained, then take the person and product to the nearest medical emergency treatment center or hospital.

Section 5	**FIRE FIGHTING MEASURES**
Flashpoint / Method:	14 - 23 Degrees F. (-10 to -5 Degrees C) / CCCFP
Flammability:	LEL = 1.8 % Volume, UEL = 11.8 % Volume
Extinguishing Media:	Use dry chemical, CO_2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure:	Firefighters should wear positive pressure self-contained breathing apparatus and full protective clothing for fires in areas where chemicals are used or stored
Unusual Fire And Explosion Hazards:	Extremely flammable liquid. Keep away from heat and all sources of ignition including sparks, flames, lighted cigarettes and pilot lights. Containers may rupture or explode in the heat of a fire. Vapors are heavier than air and may travel to a remote ignition source and flash back. This product contains tetrahydrofuran that may form explosive organic peroxide when exposed to air or light or with age.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.

Section 6	**ACCIDENTAL RELEASE MEASURES**
Spill or Leak Procedures:	Remove all sources of ignition and ventilate area. Stop leak if it can be done without risk. Personnel cleaning up the spill should wear appropriate personal protective equipment, including respirators if vapor concentrations are high. Soak up spill with an inert absorbent such as sand, earth or other non-combusting material. Put absorbent material in covered, labeled metal containers. Prevent liquid from entering watercourses, sewers and natural waterways. Report releases to authorities as required. See Section 13 for disposal information.

Section 7	**HANDLING AND STORAGE**
Handling:	Avoid contact with eyes, skin and clothing. Avoid breathing vapors or mists. Use with adequate ventilation (equivalent to outdoors). Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep product away from heat, sparks, flames and all other sources of ignition. No smoking in storage or use areas. Keep containers closed when not in use.
Storage:	Store in a cool, dry, well-ventilated area away from incompatible materials. Keep containers closed when not in use.
Other:	"Empty" containers retain product residue and can be hazardous. Follow all MSDS precautions in handling empty containers. Do not cut or weld on or near empty or full containers.

Section 8	**EXPOSURE CONTROLS/PERSONAL PROTECTION**
Ventilation:	Open doors & windows. Provide ventilation capable of maintaining emissions at the point of use below recommended exposure limits. If used in enclosed area, use exhaust fans. Exhaust fans should be explosion-proof or set up in a way that flammable concentrations of solvent vapors are not exposed to electrical

	fixtures or hot surfaces.
Respiratory Protection:	For operations where the exposure limit may be exceeded, a NIOSH approved organic vapor respirator or supplied air respirator is recommended. Equipment selection depends on contaminant type and concentration, select in accordance with 29 CFR 1910.134 and good industrial hygiene practice. For firefighting, use self-contained breathing apparatus.
Skin Protection:	Rubber gloves are suitable for normal use of the product. For long exposures chemical resistant gloves may be required such as 4H(tm) or Silver Shield(tm) to avoid prolonged skin contact.
Eye Protection:	Safety glasses with side shields or safety goggles.

Section 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	151 Degrees F / 66 Degrees C
Melting Point:	Not applicable
Vapor Pressure:	145 mmHg @ 20 Degrees C
Vapor Density:	(Air = 1) 2.5
Volatile Components:	78-82%
Solubility In Water:	Negligible
pH:	Not applicable
Specific Gravity:	0.94 +/- 0.02 @ 20 Degrees C
Evaporation Rate:	(BUAC = 1) = 5.5 - 8.0
Appearance:	Yellow/Gold Liquid
Odor:	Ether-Like
Will Dissolve In:	Tetrahydrofuran
Material Is:	Liquid

Section 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, sparks, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.
Incompatibility/ Materials To Avoid:	Oxidizing agents, alkalis, amines, ammonia, acids, chlorine compounds, chlorinated inorganics (potassium, calcium and sodium hypochlorite) and hydrogen peroxides. May attack plastic, resins and rubber.
Hazardous Polymerization:	Will not occur.

Section 11 TOXICOLOGICAL INFORMATION

Inhalation:	Vapors or mists may cause mucous membrane and respiratory irritation, coughing, headache, dizziness, dullness, nausea, shortness of breath and vomiting. High concentrations may cause central nervous system depression, narcosis and unconsciousness. May cause kidney, liver and lung damage.
Skin:	May cause irritation with redness, itching and pain. Methyl ethyl ketone and cyclohexanone may be absorbed through the skin causing effects similar to those listed under inhalation.
Eye:	Vapors may cause irritation. Direct contact may cause irritation with redness, stinging and tearing of the eyes. May cause eye damage.
Ingestion:	Swallowing may cause abdominal pain, nausea, vomiting and diarrhea. Aspiration during swallowing or vomiting can cause chemical pneumonia and lung damage. May cause kidney and liver damage.
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis and damage to the kidney, liver, lungs and central nervous system.
Toxicity Data:	Acetone: Oral rat LD50: 5,800 mg/kg

	Inhalation rat LC50: 50,100 mg/m3/8 hours
Cyclohexanone:	Oral rat LD50: 1,620 mg/kg Inhalation rat LC50: 8,000 ppm/4 hours Skin rabbit LD50: 1 mL/kg
Tetrahydrofuran:	Oral rat LD50: 1,650 mg/kg Inhalation rat LC50: 21,000 ppm/3 hours
Methyl Ethyl Ketone:	Oral rat LD50: 2,737 mg/kg Inhalation rat LC50: 23,500 mg/m3/8 hours Skin rabbit LD50: 6,480 mg/kg

Sensitization: None of the components are known to cause sensitization.

Carcinogenicity: None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA. The National Toxicology Program has reported that exposure of mice and rats to tetrahydrofuran (THF) vapor levels up to 1800 ppm 6 hr/day, 5 days/week for their lifetime caused an increased incidence of kidney tumors in male rats and liver tumors in female mice. The significance of these findings for human health is unclear at this time, and may be related to "species specific" effects. Elevated incidences of tumors in humans have not been reported for THF. ACGIH has classified cyclohexanone (CYH) and tetrahydrofuran as “A3,” Confirmed Animal Carcinogens with Unknown Relevance to Humans.

Mutagenicity: Cyclohexanone has been positive in bacterial and mammalian assays. Acetone, methyl ethyl ketone and tetrahydrofuran are generally thought not to be mutagenic.

Reproductive Toxicity: Methyl ethyl ketone and cyclohexanone have been shown to cause embryofetal toxicity and birth defects in laboratory animals. Acetone and tetrahydrofuran has been found to cause adverse developmental effects only when exposure levels cause other toxic effects to the mother.

Medical Conditions Aggravated By Exposure: Persons with pre-existing skin, lung, kidney or liver disorders may be at increased risk from exposure to this product.

Section 12 ECOLOGICAL INFORMATION

This product is not expected to be toxic to aquatic organisms.
Cyclohexanone: 96 hour LC50 values for fish is over 100 mg/l.
Tetrahydrofuran: 96 hour LC50 fathead minnow: 2160 mg/L.
Acetone: 96 hour LC50 for fish is greater than 100 mg/L.
Methyl Ethyl Ketone: 96 hour LC50 for fish is greater than 100 mg/L.

VOC Information: This product emits VOC’s (volatile organic compounds) in its use. Make sure that use of this product complies with local VOC emission regulations, where they exist.

VOC Level: Maximum 490 g/L per SCAQMD Test Method 316A.

Section 13 DISPOSAL CONSIDERATIONS

Waste Disposal:	Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number:	U002, U057, U159, U213
EPA Hazardous Waste ID Number:	D001, D035, F003, F0005
EPA Hazard Waste Number:	Ignitable Waste. Toxic Waste (Methyl Ethyl Ketone content)

Section 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)
UN/NA Number:	None	UN1133

Proper Shipping Name:	Consumer Commodity	Adhesives
Hazard Class:	ORM-D	3
Packing Group:	None	PGII
Hazard Labels:	None	Flammable Liquid
IMDG		
UN Number:	UN1133	UN1133
Proper Shipping Name:	Adhesives	Adhesives
Hazard Class:	3	3
Packing Group:	II	II
Label:	None (Limited Quantities are expected from labeling)	Class 3 (Flammable Liquid)
Flashpoint (deg C)	-10 to -5 Degrees C	-10 to -5 Degrees C

2008 North American Emercency Response Guidebook Number: 127

Section 15 REGULATORY INFORMATION

Hazard Category for Section 311/312:	Acute Health, Chronic Health, Flammable
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain chemicals subject to SARA Title III Section 313 Reporting requirements.
CERCLA 103 Reportable Quantity:	Spills of this product over the RQ (reportable quantity) must be reported to the National Response Center. The RQ for the product, based on the RQ for Tetrahydrofuran (50% maximum) of 1,000 lbs, is 2,000 lbs. Many states have more stringent release reporting requirements. Report spills required under federal, state and local regulations.
California Proposition 65:	This product contains trace amounts of chemicals known to the State of California to cause cancer. Under normal use conditions, exposure to these chemicals at levels above the State of California “No Significant Risk Level” (NSRL) are unlikely. The use of proper personal protective equipment (PPE) and ventilation guidelines noted in Section 8 will minimize exposure to these chemicals.
TSCA Inventory	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Class B, Division 2; Class D, Division 2, Subdivision B; Class D, Division 2, Subdivision A. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

Section 16 OTHER INFORMATION

NFPA and HMIS:	
NFPA Hazard Signal:	Health: 2 Flammability: 3 Reactivity: 1 Special: None
HMIS Hazard Signal:	Health: 2* Flammability: 3 Reactivity: 1 PPE: G

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, we cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

Template: tmpl-us-e1

MATERIAL SAFETY DATA SHEET

SECTION 1 PRODUCT AND COMPANY IDENTIFICATION

Trade Name: OATEY DARK THREAD CUTTING OIL
Product Use: Oil for pipe cutting and threading
Formula: Petroleum Distillates mixture
Synonyms: Thread cutting oil
Firm Name & Mailing Address: OATEY CO. 4700 West 160th Street P.O. Box 35906 Cleveland, Ohio 44135, U.S.A. http://www.oatey.com
Oatey Phone Number: (216) 267-7100 or (800) 321-9532
Emergency Phone Numbers: For Emergency First Aid call 1-303-623-5716 COLLECT. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared By: Corporate Director – Safety and Environmental Compliance
Preparation Date: July 27, 2005

SECTION 2 COMPOSITION/INFORMATION ON INGREDIENTS

INGREDIENTS:	%wt/wt:	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA:	OTHER:
Petroleum distillates	90%, max.	69029-75-0	5mg/m3	5mg/m3	STEL: 10mg/m3
Petroleum distillates	25%, max.	69029-75-0	5mg/m3	5mg/m3	STEL: 10mg/m3
Sulfur additive	10%, max.	68153-70-8	None	None	None
Chlorinated paraffin	5%, max.	63449-39-8	None	None	None

OSHA Hazard Classification: Irritant.

SECTION 3 HAZARDS IDENTIFICATION

Emergency Overview:
Transparent dark amber fluid. Mild eye, skin, inhalation and ingestion irritant.

SECTION 4 FIRST AID MEASURES

CALL 1-303-623-5716 COLLECT

Skin: Remove contaminated clothing immediately. Wash all exposed areas with soap and water. Get medical attention if irritation develops.

Eyes: If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.

Inhalation: If respiratory irritation develops, remove to fresh air. If breathing becomes difficult, administer oxygen. Administer artificial respiration if breathing has stopped. Seek immediate medical attention.

Ingestion: **DO NOT INDUCE VOMITING.** Never give anything by mouth to a person who is unconscious or drowsy. Get immediate medical attention by calling a Poison Control Center, or hospital emergency room. If medical advice cannot be obtained, then take the person and product to the nearest medical emergency treatment center or hospital.

SECTION 5 FIRE FIGHTING MEASURES

Flashpoint / Method: > 300 Degrees F (149 Degrees C) / C.O.C.

Flammability: LEL = Not available, UEL = Not available

Extinguishing Media: Use dry chemical, CO_2, or foam to extinguish fire. Do not use water as it may be ineffective as an extinguishing agent.

Special Fire Fighting Procedure: Firefighters should wear positive pressure self-contained breathing apparatus and full protective clothing for fires in areas where chemicals are used or stored

Unusual Fire and Explosion Hazards: None

Hazardous Decomposition Products: Combustion will produce toxic and irritating vapors including oxides of carbon, phosphorus, sulfur, smoke, hydrogen chloride, and hydrogen sulfide.

SECTION 6 ACCIDENTAL RELEASE MEASURES

Spill or Leak Procedures: Remove all sources of ignition and ventilate area. Contain spill, stop source of leak, pump spilled material into salvage container, soak up remaining material with absorbent such as sand or clay. Prevent product from entering potable or natural water systems. Put absorbent material in covered, labeled metal containers. Report releases to authorities as required. See Section 13 for disposal information.

SECTION 7 HANDLING AND STORAGE

Handling: Avoid prolonged and repeated skin contact. Avoid contact with eyes, skin and clothing. Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep product away from heat, sparks, flames and all other sources of ignition. No smoking in storage or use areas. Keep containers closed when not in use. Launder contaminated clothing before reuse.

Storage: Store in a cool, dry, well-ventilated area away from incompatible materials. Store away from direct heat source and strong oxidizing agents. Keep containers closed when not in use.

Other: Do not pressurize, cut, weld, braze, drill, grind or heat empty containers as some flammable, hazardous or combustible residue may be present. Avoid breathing mist. For industrial use only.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Ventilation: General mechanical ventilation is usually adequate under normal conditions of use. Use local ventilation if necessary to maintain oil mist below PEL/TLV.

Respiratory Protection: For operations where the exposure limit may be exceeded, a NIOSH approved organic vapor respirator or supplied air respirator is recommended. Equipment selection depends on contaminant type and concentration, select in accordance with 29 CFR 1910.134 and good industrial hygiene practice.

Skin Protection: Oil/Chemical resistant gloves such as neoprene or nitrile.

Eye Protection: Safety glasses with side shields or chemical splash goggles if necessary.

SECTION 8: Continued

Other: Eye wash and safety shower should be available. Other such as protective apron as necessary to prevent prolonged skin contact.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	> 500 Degrees F / 260 Degrees C
Melting Point:	Not applicable
Vapor Pressure:	< 0.1 mm Hg at 20 degrees C
Vapor Density:	> 1.0 (Air = 1.0)
Volatile Components:	Negligible
Solubility In Water:	Negligible
pH:	Not applicable
Specific Gravity:	0.90 – 0.95 (Water = 1.0)
Evaporation Rate:	Negligible (BUAC = 1.0)
Appearance:	Transparent dark amber oily fluid
Odor:	Mild petroleum
Will Dissolve In:	Oil and organic solvents
Material Is:	Liquid

SECTION 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, sparks, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including oxides of carbon, phosphorus, sulfur, smoke, hydrogen chloride, and hydrogen sulfide.
Incompatibility/ Materials To Avoid:	Avoid contact with strong oxidizers.
Hazardous Polymerization:	Will not occur.

SECTION 11 TOXICOLOGICAL INFORMATION

Inhalation:	Excessive exposure to mist may irritate respiratory tract.
Skin:	May cause irritation.
Eye:	May cause irritation.
Ingestion:	May irritate gastrointestinal tract.
Chronic Toxicity:	Prolonged continued skin contact may cause dermatitis, mist may irritate respiratory tract.
Toxicity Data:	Not available
Sensitization:	None of the components are known to cause sensitization.
Carcinogenicity:	None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA.
Mutagenicity:	Not available
Reproductive Toxicity:	Not available
Medical Conditions Aggravated By Exposure:	Persons with pre-existing skin or lung disorders may be at increased risk from exposure to this product.

SECTION 12 ECOLOGICAL INFORMATION

VOC
Information: None
VOC Level: Not established

SECTION 13 DISPOSAL CONSIDERATIONS

Waste Disposal: Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number: Not applicable
EPA Hazardous Waste ID Number: Not applicable
EPA Hazard Waste Class: Not applicable

SECTION 14 TRANSPORT INFORMATION

DOT

Proper Shipping Name:	Not applicable
Hazard Class/Packing Group:	Not applicable
UN/NA Number:	Not applicable
Hazard Labels:	Not applicable

IMDG

Proper Shipping Name:	Not applicable
Hazard Class/Packing Group:	Not applicable
UN Number:	Not applicable
Label:	Not applicable

2004 North American Emergency Response Guidebook Number: None

SECTION 15 REGULATORY INFORMATION

Hazard Category for Section 311/312:	None
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product contains no chemicals subject to SARA Title III Section 313 Reporting requirements.
CERCLA 103 Reportable:	None
California Proposition 65:	This product is not known to contain any chemicals subject to California Proposition 65 regulation.
TSCA Inventory:	All of the components of this product are listed on the TSCA inventory.
Canadian WHMIS Classification:	D2B. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

SECTION 16 OTHER INFORMATION

NFPA and HMIS:
NFPA Hazard Signal: Health: 1 Flammability: 1 Reactivity: 0 Special: None
HMIS Hazard Signal: Health: 1 Flammability: 1 Reactivity: 0 PPE: B

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, Oatey cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

MATERIAL SAFETY DATA SHEET

MSDS Name: POLYSEAMSEAL TUB & TILE Adhesive Caulk
MSDS Number: sD11515
Version Number 1
MSDS Date: JAN-24-2001
Page Number: 1

SECTION I - PRODUCT AND COMPANY INFORMATION

Product Name:	POLYSEAMSEAL TUB & TILE Adhesive Caulk
Hazard Rating:	Health: 1 Fire: 0 Reactivity: 0 PPI: B
Company Identification:	OSI SEALANTS, INC. 7405 PRODUCTION DRIVE MENTOR OH 44060
Contact:	Safety Officer
Telephone/Fax:	(440) 255-8900 (440) 974-2395
Emergency Phone (24 hour)	CHEMTREC (800) 424-9300
Chemtrec (outside-USA)	(703) 527-3887
Preparer	T.F.Barr Sr. R.&D. Chemist
Product Class	CAULK
Trade Name	POLYSEAMSEAL
Product Code	---

SECTION II - INGREDIENT AND HAZARD INFORMATION

Ingredient Name	CAS Number	Percent	TSCA
N-Butyl Acetate	0123-86-4	< 5.0	Y
Titanium Dioxide **	13463-67-7	< 2.0	Y

Ingredient Notes:
**The main hazard for Titanium Dioxide is dust inhalation. Because of its incorporation into the caulk, exposure by inhalation is unlikely.

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SECTION III - PHYSICAL AND CHEMICAL PROPERTIES

Form:	High viscosity caulk
Appearance/Color:	White & colors
Odor:	Banana odor
Solubility (in water):	yes
pH Value, +/-.3:	7.5
Boiling Range:	212.øF (100.øC)
Vapor Pressure (mmHg):	15.@ 68.øF (20.øC)

Evaporation Rate:	0.5 times Slower than n-Butyl Acetate
Vapor Density:	Heavier than air
% Volatile, Weight	approx. 30.%
% Volatile, Volume	approx. 45.%
Specific Gravity:	1.245
VOC (less H2O or exempt)	93 Gr/L
Heavy Elements (ppm)	0.

NOTE:
Freeze Point : 32 Deg F

- -

SECTION IV - FIRE FIGHTING MEASURES (Flash,UEL,LEL for solvent only)

Flammability Class	N/A
Flash Range:	Not Available
Explosive Range (LEL/UEL):	Not Applicable

EXTINGUISHING MEDIA:
Dry chemical -- Carbon Dioxide -- Foam -- Water Fog
Will not burn in wet state.
SPECIAL FIRE-FIGHTING PROCEDURES:
Water may be used to cool and protect exposed containers.
Caution should be taken because uncured material is water soluble.
UNUSUAL FIRE AND EXPLOSION HAZARDS:
Any closed container may burst when exposed to extreme
heat or fire.

- -

SECTION V - HEALTH HAZARD DATA

ROUTES OF ENTRY:
ENTRY THROUGH...
Inhalation? Yes | Skin? Yes | Ingestion? Yes
TARGET ORGANS... --Butyl Acetate: Eyes
-- Titanium Dioxide: Eyes, Skin, Lung
Overexposure -- Butyl Acetate: Eye contact causes severe
irritation. Vapors may be irritating and cause
dizziness, headache, nausea, vomiting and narcosis.
-- Titanium Dioxide: Respiratory Tract Irritant(Dust only)
CARCINOGENICITY...NTP? N/E| IARC NO | OSHA? N/E
EFFECTS OF OVEREXPOSURE

Inhalation: May irritate respiratory tract.
Skin/Eye Contact: May cause irritation.
Ingestion: Swallowing large amounts may cause nausea, vomiting. (an unlikely route of entry)

NOTICE: Reports have associated repeated and prolonged occupational overexposure to solvents with permanent brain, nervous system, liver or kidney damage or may cause cardiac arrhythmia. INTENTIONAL misuse of this product by deliberately inhaling its vapors may be harmful or fatal.

FIRST AID MEASURES

Inhalation: If affected by inhalation, remove to fresh air.
Eye Contact: Flush with water for at least 15 minutes, and get prompt medical attention.
Skin Contact: Wash skin thoroughly with soap and water.
Ingestion: Drink water, and get medical attention.

N/A= Not applicable N/AV=Not available N/E, N/est=Not established

- -

SECTION VI - STABILITY AND REACTIVITY

Stability: This product is stable
Hazardous Polymerization: Hazardous polymerization will not occur

INCOMPATABILITY:
None
CONDITIONS TO AVOID:
None
HAZARDOUS DECOMPOSITION PRODUCTS:
May produce oxides of carbon and oxides of nitrogen when burned.

- -

SECTION VII - ACCIDENTAL RELEASE AND DISPOSAL MEASURES:

STEPS TO BE TAKEN IN CASE OF SPILL:
Wear appropriate protective clothing. Add dry absorbent and shovel or sweep up. Place in an appropriate container and seal.
WASTE DISPOSAL METHOD:
Dispose of in accordance with Federal, State, and Local regulations.

- -

SECTION VIII - EXPOSURE CONTROLS/PERSONAL PROTECTION

Occupational Exposure Limits

	ACGIH TLV	ACGIH TLV-C	ACGIH STEL	OSHA STEL	OSHA PEL

N-Butyl Acetate	150.00 PPM	N/est	200.00 PPM	200.00 PPM	150.00 PPM
Titanium Dioxide **	10.00 mg/F3	N/est	N/est	N/est	15.00 mg/F3

RESPIRATORY PROTECTION:
NIOSH respirators recommended if vapors and mists are generated.
VENTILATION:
Local exhaust is recommended for safe practice.
PROTECTIVE CLOTHING:
Rubber gloves and impervious clothing should be worn to prevent repeated skin contact.
EYE PROTECTION:
Splashproof goggles or safety glasses should be worn.
HANDLING AND STORAGE PRECAUTIONS:
Keep from freezing. Keep away from heat.
Keep out of the reach of children
Keep containers closed when not in use
Avoid prolonged or repeated contact with skin.

- -

SECTION IX - TRANSPORT INFORMATION:
DOT CLASSIFICATIONS FOR ONE GALLON AND UNDER CONTAINERS
DESCRIPTION: Non-hazardous
DOT CLASSIFICATIONS for OVER 1 GALLON size CONTAINERS
DESCRIPTION: Non-hazardous
Caulking type product

- -

SECTION X - REGULATORY INFORMATION:
The following ingredients are registered for TSCA 12B

Ingredient Name	CAS Number	Percent
N-Butyl Acetate	0123-86-4	4.76

The components fit the EPA Hazard Category definition of Immediate (Acute) and Delayed (Chronic) Health Hazards under SARA Sections 311, 312.
All chemical substances are TSCA listed.
SARA Section 313 toxic chemicals: NONE above the de minimis reporting thresholds.
California PROP.65 Chemicals: none known

DISCLAIMER:
The information contained herein is based on data available as of the date of preparation of this MSDS and which we believe to be reliable. However, no warranty is expressed or implied regarding the accuracy of the data. We shall not be responsible for the use of this information, or of any product, method, or apparatus mentioned, and the user must make his own investigation to determine the suitability of the information or products for his particular purpose, for the protection of the environment, and the health and safety of the users of this material.

MATERIAL SAFETY DATA SHEET

MSDS Name: POLYSEAMSEAL TUB & TILE Adhesive Caulk - CLEAR
MSDS Number: sD11515C
Version Number
MSDS Date: JAN-24-2001
Page Number: 1

SECTION I - PRODUCT AND COMPANY INFORMATION

Product Name:	POLYSEAMSEAL TUB & TILE Adhesive Caulk - CLEAR
Hazard Rating:	Health: 1 Fire: 0 Reactivity: 0 PPI: B
Company Identification:	OSI SEALANTS, INC. 7405 PRODUCTION DRIVE MENTOR OH 44060
Contact:	Safety Officer
Telephone/Fax:	(440) 255-8900 (440) 974-2395
Emergency Phone (24 hour)	CHEMTREC (800) 424-9300
Chemtrec (outside-USA)	(703) 527-3887
Preparer	T.F.Barr Sr. R.&D. Chemist
Product Class	CAULK
Trade Name	POLYSEAMSEAL
Product Code	-- Clear

SECTION II - INGREDIENT AND HAZARD INFORMATION

Ingredient Name	CAS Number	Percent	TSCA

No OSHA hazardous ingredients known at this time.

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SECTION III - PHYSICAL AND CHEMICAL PROPERTIES

Form:	Milky-white caulk
Appearance/Color:	CLEAR when dry
Solubility (in water):	Yes
pH Value, +/-.3:	7.5
Boiling Range:	212.øF (100.øC)
Vapor Pressure (mmHg):	15.@ 68.øF (20.øC)
Evaporation Rate:	0.5 times Slower than n-Butyl Acetate
Vapor Density:	Heavier than air
% Volatile, Weight	38.%

% Volatile, Volume 40.%
Specific Gravity: 1.06
VOC (less H2O or exempt) <15 g/l
Heavy Elements (ppm) 0.

NOTE:
Odor : Mild acrylic odor. Freeze Point : 32 Deg F

- -

SECTION IV - FIRE FIGHTING MEASURES (Flash,UEL,LEL for solvent only)
Flammability Class N/A
Flash Range: Not Applicable
Explosive Range (LEL/UEL): Not Applicable

EXTINGUISHING MEDIA:
Dry chemical -- Carbon Dioxide -- Foam -- Water Fog
Will not burn in wet state.
SPECIAL FIRE-FIGHTING PROCEDURES:
Water may be used to cool and protect exposed containers.
Caution should be taken because uncured material is water soluble.
UNUSUAL FIRE AND EXPLOSION HAZARDS:
Any closed container may burst when exposed to extreme
heat or fire.

- -

SECTION V - HEALTH HAZARD DATA

ROUTES OF ENTRY:
Inhalation? Yes | Skin? Yes | Ingestion? Yes
TARGET ORGANS...
None- No hazardous ingredients per OSHA Regulations
CARCINOGENICITY...
NTP? N/A | IARC Monographs? N/A | OHSA? NO
EFFECTS OF OVEREXPOSURE
Inhalation: May irritate respiratory tract.
Skin/Eye Contact: May cause irritation.
Ingestion: Swallowing large amounts may cause nausea,
vomiting. (an unlikely route of entry)
FIRST AID MEASURES
Inhalation: If affected by inhalation, remove to fresh air.
Eye Contact: Flush with water for at least 15 minutes, and
get prompt medical attention.
Skin Contact: Wash skin thoroughly with soap and water.

Ingestion: Drink water, and get medical attention.

N/A= Not applicable N/AV=Not available N/E, N/est=Not established

- -

SECTION VI - STABILITY AND REACTIVITY

Stability: This product is stable

Hazardous Polymerization: Hazardous polymerization will not occur

INCOMPATABILITY:
None

CONDITIONS TO AVOID:
None

HAZARDOUS DECOMPOSITION PRODUCTS:
May produce oxides of carbon and oxides of nitrogen when burned.

- -

SECTION VII - ACCIDENTAL RELEASE AND DISPOSAL MEASURES:

STEPS TO BE TAKEN IN CASE OF SPILL:
Wear appropriate protective clothing. Add dry absorbent and shovel or sweep up. Place in an appropriate container and seal.

WASTE DISPOSAL METHOD:
Dispose of in accordance with Federal, State, and Local regulations.

- -

SECTION VIII - EXPOSURE CONTROLS/PERSONAL PROTECTION

Occupational Exposure Limits

	ACGIH TLV	ACGIH TLV-C	ACGIH STEL	OSHA STEL	OSHA PEL

RESPIRATORY PROTECTION:
NIOSH respirators recommended if vapors and mists are generated.

VENTILATION:
Local exhaust is recommended for safe practice.

PROTECTIVE CLOTHING:
Rubber gloves and impervious clothing should be worn to prevent repeated skin contact.

EYE PROTECTION:
Splashproof goggles or safety glasses should be worn.

HANDLING AND STORAGE PRECAUTIONS:
Keep from freezing. Keep away from heat.
Keep out of the reach of children
Keep containers closed when not in use

Avoid prolonged or repeated contact with skin.

- -

SECTION IX - TRANSPORT INFORMATION:

DOT CLASSIFICATIONS FOR ONE GALLON AND UNDER CONTAINERS

DESCRIPTION: Non-hazardous
DOT CLASSIFICATIONS for OVER 1 GALLON size CONTAINERS

DESCRIPTION: Non-hazardous
Caulking type product

- -

SECTION X - REGULATORY INFORMATION:

This product is regulated by SARA Sections 311,312 solely because it is regulated by OSHA Hazard Communication standard (29CFR 1910.1200). However none of the five Hazard Catagories (SARA 311, 312) are appropriate for this product.
SARA Section 313 toxic chemicals: NONE
All ingredients TSCA listed.
California PROP.65 Chemicals: none known

DISCLAIMER:
The information contained herein is based on data available as of the date of preparation of this MSDS and which we believe to be reliable. However, no warranty is expressed or implied regarding the accuracy of the data. We shall not be responsible for the use of this information, or of any product, method, or apparatus mentioned, and the user must make his own investigation to determine the suitability of the information or products for his particular purpose, for the protection of the environment, and the health and safety of the users of this material.

MATERIAL SAFETY DATA SHEET

MSDS Number: 1106E

Section 1 **PRODUCT AND COMPANY IDENTIFICATION**

Trade Name: OATEY ALL PURPOSE CEMENT
Product Nos.: Milky - 30818, 30821, 30834, 30847, 30848, 31650, 31651, 32208, 32209
Product Use: Cement for PVC, CPVC and ABS pipe
Formula: PVC and CPVC Resin in Solvent Solution
Synonyms: None
Firm Name & Address: Oatey Company 4700 West 160th Street, Cleveland, Ohio 44135 www.oatey.com
Firm Phone No: (216) 267-7100
Emergency Phone Nos.: For Emergency First Aid call 1-877-740-5015. For chemical transportation emergencies ONLY, call Chemtrec at 1-800-424-9300. Outside the U.S. 1-703-527-3887.
Prepared by: Technical Department
Preparation Date: 10/01/2009

Section 2 **HAZARDS IDENTIFICATION**

Emergency Overview:
Milky liquid with an ether-like odor. Extremely flammable liquid and vapor. Vapors may cause flash fire. May cause eye and skin irritation. Inhalation of vapors or mist may cause respiratory irritation and central nervous system effects. Swallowing may cause irritation, nausea, vomiting, diarrhea and kidney or liver disorders. Aspiration hazard. May be fatal if swallowed. Symptoms may be delayed.

Section 3 **COMPOSITION/INFORMATION ON INGREDIENTS**

INGREDIENTS:	%wt/wt :	CAS NUMBER:	ACGIH TLV TWA:	OSHA PEL TWA	OTHER:
Tetrahydrofuran	30 - 45%	109-99-9	50 ppm(skin) 100 ppm STEL	200 ppm	25 ppm (Mfg)
Methyl Ethyl Ketone	8 - 18%	78-93-3	200 ppm 300 ppm STEL	200 ppm	None
Acetone	10 – 20%	67-64-1	500 ppm 750 ppm STEL	1000 ppm	None
PVC Resin (Non-hazardous)	8 – 15%	9002-86-2	10 mg/m3	15 mg/m3	None
CPVC Resin (Non-hazardous)	3 – 7%	68648-82-8	10 mg/m3	None Established	None
Cyclohexanone	10 – 20%	108-94-1	20 ppm(skin) 50 ppm STEL	50 ppm	None
Amorphous Fumed Silica (Non-hazardous)	1 - 5%	112945-52-5	10 mg/m3	None Established	None a

OSHA Hazard Classification: Flammable, irritant, organ effects

Section 4 **FIRST AID MEASURES**

Skin: Remove contaminated clothing immediately. Wash all exposed areas with soap and

	water. Get medical attention if irritation develops. Remove dried cement with hand cleaner or baby oil.
Eyes:	If material gets into eyes or if fumes cause irritation, immediately flush eyes with plenty of water until chemical is removed. If irritation persists, get medical attention immediately.
Inhalation:	If symptoms of exposure develop, remove to fresh air. If breathing becomes difficult, administer oxygen. Administer artificial respiration if breathing has stopped. Seek immediate medical attention.
Ingestion:	**DO NOT INDUCE VOMITING.** Rinse mouth with water. Never give anything by mouth to a person who is unconscious or drowsy. Get immediate medical attention by calling a Poison Control Center, or hospital emergency room. If medical advice cannot be obtained, then take the person and product to the nearest medical emergency treatment center or hospital.

Section 5	**FIRE FIGHTING MEASURES**
Flashpoint / Method:	14 - 23 Degrees F. (-10 to -5 Degrees C) / CCCFP
Flammability:	LEL = 1.8 % Volume, UEL = 11.8 % Volume
Extinguishing Media:	Use dry chemical, CO_2, or foam to extinguish fire. Cool fire exposed container with water. Water may be ineffective as an extinguishing agent.
Special Fire Fighting Procedure:	Firefighters should wear positive pressure self-contained breathing apparatus and full protective clothing for fires in areas where chemicals are used or stored
Unusual Fire And Explosion Hazards:	Extremely flammable liquid. Keep away from heat and all sources of ignition including sparks, flames, lighted cigarettes and pilot lights. Containers may rupture or explode in the heat of a fire. Vapors are heavier than air and may travel to a remote ignition source and flash back. This product contains tetrahydrofuran that may form explosive organic peroxide when exposed to air or light or with age.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.

Section 6	**ACCIDENTAL RELEASE MEASURES**
Spill or Leak Procedures:	Remove all sources of ignition and ventilate area. Stop leak if it can be done without risk. Personnel cleaning up the spill should wear appropriate personal protective equipment, including respirators if vapor concentrations are high. Soak up spill with an inert absorbent such as sand, earth or other non-combusting material. Put absorbent material in covered, labeled metal containers. Prevent liquid from entering watercourses, sewers and natural waterways. Report releases to authorities as required. See Section 13 for disposal information.

Section 7	**HANDLING AND STORAGE**
Handling:	Avoid contact with eyes, skin and clothing. Avoid breathing vapors or mists. Use with adequate ventilation (equivalent to outdoors). Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep product away from heat, sparks, flames and all other sources of ignition. No smoking in storage or use areas. Keep containers closed when not in use.
Storage:	Store in a cool, dry, well-ventilated area away from incompatible materials. Keep containers closed when not in use.
Other:	"Empty" containers retain product residue and can be hazardous. Follow all MSDS precautions in handling empty containers. Do not cut or weld on or near empty or full containers.

Section 8	**EXPOSURE CONTROLS/PERSONAL PROTECTION**
Ventilation:	Open doors & windows. Provide ventilation capable of maintaining emissions at the point of use below recommended exposure limits. If used in enclosed area, use exhaust fans. Exhaust fans should be explosion-proof or set up in a way

	that flammable concentrations of solvent vapors are not exposed to electrical fixtures or hot surfaces.
Respiratory Protection:	For operations where the exposure limit may be exceeded, a NIOSH approved organic vapor respirator or supplied air respirator is recommended. Equipment selection depends on contaminant type and concentration, select in accordance with 29 CFR 1910.134 and good industrial hygiene practice. For firefighting, use self-contained breathing apparatus.
Skin Protection:	Rubber gloves are suitable for normal use of the product. For long exposures chemical resistant gloves may be required such as 4H(tm) or Silver Shield(tm) to avoid prolonged skin contact.
Eye Protection:	Safety glasses with side shields or safety goggles.

Section 9 PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point:	151 Degrees F / 66 Degrees C
Melting Point:	Not applicable
Vapor Pressure:	145 mmHg @ 20 Degrees C
Vapor Density:	(Air = 1) 2.5
Volatile Components:	80-84%
Solubility In Water:	Negligible
pH:	Not applicable
Specific Gravity:	0.94 +/- 0.02 @ 20 Degrees C
Evaporation Rate:	(BUAC = 1) = 5.5 - 8.0
Appearance:	Milky Liquid
Odor:	Ether-Like
Will Dissolve In:	Tetrahydrofuran
Material Is:	Liquid

Section 10 STABILITY AND REACTIVITY

Stability:	Stable.
Conditions To Avoid:	Avoid heat, sparks, flames and other sources of ignition.
Hazardous Decomposition Products:	Combustion will produce toxic and irritating vapors including carbon monoxide, carbon dioxide and hydrogen chloride.
Incompatibility/ Materials To Avoid:	Oxidizing agents, alkalis, amines, ammonia, acids, chlorine compounds, chlorinated inorganics (potassium, calcium and sodium hypochlorite) and hydrogen peroxides. May attack plastic, resins and rubber.
Hazardous Polymerization:	Will not occur.

Section 11 TOXICOLOGICAL INFORMATION

Inhalation:	Vapors or mists may cause mucous membrane and respiratory irritation, coughing, headache, dizziness, dullness, nausea, shortness of breath and vomiting. High concentrations may cause central nervous system depression, narcosis and unconsciousness. May cause kidney, liver and lung damage.
Skin:	May cause irritation with redness, itching and pain. Methyl ethyl ketone and cyclohexanone may be absorbed through the skin causing effects similar to those listed under inhalation.
Eye:	Vapors may cause irritation. Direct contact may cause irritation with redness, stinging and tearing of the eyes. May cause eye damage.
Ingestion:	Swallowing may cause abdominal pain, nausea, vomiting and diarrhea. Aspiration during swallowing or vomiting can cause chemical pneumonia and lung damage. May cause kidney and liver damage.
Chronic Toxicity:	Prolonged or repeated overexposure cause dermatitis and damage to the kidney, liver, lungs and central nervous system.

Toxicity Data:	Acetone:	Oral rat LD50: 5,800 mg/kg Inhalation rat LC50: 50,100 mg/m3/8 hours
	Cyclohexanone:	Oral rat LD50: 1,620 mg/kg Inhalation rat LC50: 8,000 ppm/4 hours Skin rabbit LD50: 1 mL/kg
	Tetrahydrofuran:	Oral rat LD50: 1,650 mg/kg Inhalation rat LC50: 21,000 ppm/3 hours
	Methyl Ethyl Ketone:	Oral rat LD50: 2,737 mg/kg Inhalation rat LC50: 23,500 mg/m3/8 hours Skin rabbit LD50: 6,480 mg/kg

Sensitization: None of the components are known to cause sensitization.

Carcinogenicity: None of the components are listed as a carcinogen or suspect carcinogen by NTP, IARC or OSHA. The National Toxicology Program has reported that exposure of mice and rats to tetrahydrofuran (THF) vapor levels up to 1800 ppm 6 hr/day, 5 days/week for their lifetime caused an increased incidence of kidney tumors in male rats and liver tumors in female mice. The significance of these findings for human health is unclear at this time, and may be related to "species specific" effects. Elevated incidences of tumors in humans have not been reported for THF. ACGIH has classified cyclohexanone (CYH) and tetrahydrofuran as “A3,” Confirmed Animal Carcinogens with Unknown Relevance to Humans.

Mutagenicity: Cyclohexanone has been positive in bacterial and mammalian assays. Acetone, methyl ethyl ketone and tetrahydrofuran are generally thought not to be mutagenic.

Reproductive Toxicity: Methyl ethyl ketone and cyclohexanone have been shown to cause embryofetal toxicity and birth defects in laboratory animals. Acetone and tetrahydrofuran has been found to cause adverse developmental effects only when exposure levels cause other toxic effects to the mother.

Medical Conditions Aggravated By Exposure: Persons with pre-existing skin, lung, kidney or liver disorders may be at increased risk from exposure to this product.

Section 12 ECOLOGICAL INFORMATION

This product is not expected to be toxic to aquatic organisms.
Cyclohexanone: 96 hour LC50 values for fish is over 100 mg/l.
Tetrahydrofuran: 96 hour LC50 fathead minnow: 2160 mg/L.
Acetone: 96 hour LC50 for fish is greater than 100 mg/L.
Methyl Ethyl Ketone: 96 hour LC50 for fish is greater than 100 mg/L.

VOC Information: This product emits VOC’s (volatile organic compounds) in its use. Make sure that use of this product complies with local VOC emission regulations, where they exist.

VOC Level: Maximum 510 g/L per SCAQMD Test Method 316A.

Section 13 DISPOSAL CONSIDERATIONS

Waste Disposal:	Dispose in accordance with current local, state and federal regulations.
RCRA Hazardous Waste Number:	U002, U057, U159, U213
EPA Hazardous Waste ID Number:	D001, D035, F003, F0005
EPA Hazard Waste Number:	Ignitable Waste. Toxic Waste (Methyl Ethyl Ketone content)

Section 14 TRANSPORT INFORMATION

DOT	Less than 1 Liter (0.3 gal)	Greater than 1 Liter (0.3 gal)

UN/NA Number:	None	UN1133
Proper Shipping Name:	Consumer Commodity	Adhesives
Hazard Class:	ORM-D	3
Packing Group:	None	PGII
Hazard Labels:	None	Flammable Liquid
IMDG		
UN Number:	UN1133	UN1133
Proper Shipping Name:	Adhesives	Adhesives
Hazard Class:	3	3
Packing Group:	II	II
Label:	None (Limited Quantities are expected from labeling)	Class 3 (Flammable Liquid)
Flashpoint (deg C)	-10 to -5 Degrees C	-10 to -5 Degrees C

2008 North American Emercency Response Guidebook Number: 127

Section 15 REGULATORY INFORMATION

Hazard Category for Section 311/312:	Acute Health, Chronic Health, Flammable
Section 302 Extremely Hazardous Substances (TPQ):	This product does not contain chemicals regulated under SARA Section 302.
Section 313 Toxic Chemicals:	This product does not contain chemicals subject to SARA Title III Section 313 Reporting requirements.
CERCLA 103 Reportable Quantity:	Spills of this product over the RQ (reportable quantity) must be reported to the National Response Center. The RQ for the product, based on the RQ for Tetrahydrofuran (45% maximum) of 1,000 lbs, is 2,222 lbs. Many states have more stringent release reporting requirements. Report spills required under federal, state and local regulations.
California Proposition 65:	This product contains trace amounts of chemicals known to the State of California to cause cancer. Under normal use conditions, exposure to these chemicals at levels above the State of California “No Significant Risk Level” (NSRL) are unlikely. The use of proper personal protective equipment (PPE) and ventilation guidelines noted in Section 8 will minimize exposure to these chemicals.
TSCA Inventory	All of the components of this product are listed on the TSCA inventory.
Canadian WHIMS Classification:	Class B, Division 2; Class D, Division 2, Subdivision B; Class D, Division 2, Subdivision A. This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.

Section 16 OTHER INFORMATION

NFPA and HMIS:	
NFPA Hazard Signal:	Health: 2 Flammability: 3 Reactivity: 1 Special: None
HMIS Hazard Signal:	Health: 2* Flammability: 3 Reactivity: 1 PPE: G

Disclaimer:
The information herein has been compiled from sources believed to be reliable, up-to-date, and is accurate to the best of our knowledge. However, we cannot give any guarantees regarding information from other sources, and expressly does not make warranties, nor assumes any liability for its use.

Template: tmpl-us-e1

MATERIAL SAFETY DATA SHEET **CODE: M/L 040**

This Material Safety Data Sheet complies with the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200

PRODUCT:
LENOX® STERLING® LEAD-FREE SOLDER

NFPA/HMIS HAZARD CODES: HEALTH: 1/1 FIRE: 0/0 REACTIVITY: 0/0 SPECIAL: NA

0 = Minimal 1 = Slight 2 = Moderate 3 = Serious 4 = Severe

SECTION I

MANUFACTURER NAME: LENOX®
1690 Lowery Street
Winston-Salem, NC 27101

ISSUE DATE: October, 2007

INFORMATION PHONE: 336-777-8600

SECTION II HAZARDOUS INGREDIENTS

INGREDIENT	CAS NO.	US-NIOSH RTECS NO.	US OSHA AL	US OSHA PEL	ACGIH TLV	WT. PERCENT
Tin	7440-31-5	XP7320000	NE	2.0mg/m3	2.0mg/m3	Balance
Copper	7440-50-8 (dust)	GL5325000	NE	1.0mg/m3	1.0mg/m3	4.0 – 5.0
	(fume)			0.1mg/m3	0.2mg/m3	
Selenium	7782-49-2	VS7700000	NE	0.2mg/m3	0.2mg/m3	.04 - .2

NE = NONE ESTABLISHED AL = ACTION LEVEL PEL = PERMISSIBLE EXPOSURE LIMIT TLV = THRESHOLD LIMIT VALUE

SECTION III PHYSICAL DATA

APPEARANCE & ODOR (AT NORMAL CONDITIONS): Solid - silver to silver-gray metallic metal - no odor
SPECIFIC GRAVITY: 7.38
MELTING POINT RANGE (DEGREES F): 410-418
BOILING POINT (DEGREES C): Information not available
SOLUBILITY IN WATER: Insoluble
PH: Not applicable

SECTION IV FIRE & EXPLOSION HAZARD DATA

FLASH POINT: Non-flammable
FLAMMABLE LIMITS: Not applicable
EXTINGUISHING MEDIA: No specific agents recommended
SPECIAL FIRE FIGHTING PROCEDURES: If involved in fire, use full protective clothing and NIOSH/MSHA approved self-contained breathing apparatus operated in a positive-pressure mode.
UNUSUAL FIRE & EXPLOSION HAZARDS: The solid metal form is not a fire hazard. However, it is possible that dust generated from processing operations may present a moderate fire or explosion hazard.

SECTION V REACTIVITY DATA

STABILITY: Stable
CONDITIONS TO AVOID: Not applicable
INCOMPATIBILITY: Chlorine, Turpentine, Magnesium, and Acetylene Gas
HAZARDOUS DECOMPOSITION PRODUCTS: At temperatures above the melting point metal oxide fumes may be evolved.
HAZARDOUS POLYMERIZATION: Will not occur.

SECTION VI HEALTH HAZARD DATA

NOTE: Exposure to the solid form of this product presents few health hazards in itself. However, normal handling or processing of this material may result in exposure to product components and/or decomposition products, which may present a health hazard.

ROUTES OF ENTRY:	Dust/fume inhalation; dust ingestion.
SYMPTOMS & EFFECTS OF OVEREXPOSURE:	***Chronic (prolonged)*** overexposure to **Tin** can result in benign pneumoconiosis (stannous). This form of pneumoconiosis produces progressive x-ray changes of the lungs as long as exposure exists, but there is no distinctive fibrosis, no evidence of disability and no special complicating factors. ***Acute (severe short-term)*** overexposure to **Tin** can cause irritation of the eyes, skin, mucous membranes and respiratory system. Acute overexposure to **Copper** dust/fume can cause irritation of the eyes, nose, throat and skin, and under severe fume overexposure can cause metal fume fever with flu-like symptoms such as sweet metal taste, dry throat, coughing, fever and chills, tight chest, dyspnea, headache, blurred vision, back pain, nausea, vomiting, fatigue. Symptoms usually disappear within 24 hours. **Copper** may cause skin and hair discoloration. Inhalation of copper dusts may cause changes in the gums and mucous lining of the mouth which is generally attributable to localized tissue effect rather than general toxicity.
MEDICAL CONDITIONS POSSIBLY AGGRAVATED BY EXPOSURE:	Pre-existing conditions of the lungs, Wilson's Disease (genetic trait)
CARCINOGENITY:	Not listed as a carcinogen by NTP, IARC, OSHA, ACGIH.
EMERGENCY & FIRST AID PROCEDURES:	
SKIN:	Normal hygiene and first aid procedures - wash with soap and water. If irritation develops or persists obtain medical attention get medical attention.
EYES:	Flush well with running water to remove particulate. If irritation persists obtain medical attention.
ACUTE INHALATION:	Remove from exposure. Obtain immediate medical attention. If breathing has stopped, initiate artificial resuscitation.
INGESTION:	Give water; induce vomiting only in a conscious non-convulsing individual; Obtain immediate medical attention.

SECTION VII PROTECTION MEASURES

RESPIRATORY PROTECTION:	Respiratory protection is required where airborne exposures exceed U.S. OSHA/ACGIH permissible air concentrations. Respirator selection shall be made in accordance with the U.S. OSHA Respiratory Protection Standard, 29 CFR 1910.134.
VENTILATION:	Ventilation, as described in "Industrial Ventilation, A Manual of Recommended Practice," by the American Conference of Governmental Industrial Hygienists, is recommended to maintain exposure levels below the Permissible Exposure Limits (PEL's) or Threshold Limit Values (TLV's) specified by U.S. OSHA or other local or state regulations.
PROTECTIVE GLOVES:	Recommended for prolonged contact/heat.
EYE PROTECTION:	Safety glasses or goggles are recommended where the possibility exists of getting dust particles in the eyes. Safety glasses or goggles with face shield are recommended around molten metal.
OTHER PROTECTIVE EQUIPMENT:	Safety equipment should be worn as appropriate for the work environment.
WORK/HYGIENIC PRACTICES:	Do not permit eating, drinking, or the use of cosmetics or tobacco products while handling or processing material or in product work areas. Practice good personal hygiene procedures. Wash hands and face thoroughly before eating, drinking, applying cosmetics or using tobacco products. Avoid inhalation and ingestion of product, and activities, which generate dust or fume. Keep melting/soldering temperatures as low as possible to minimize the generation of fume.

SECTION VIII PRECAUTIONS FOR SAFE HANDLING & USE

PRECAUTIONS TO BE TAKEN IN HANDLING & STORING: Practice good housekeeping procedures to prevent dust accumulations. Keep material dry. Avoid storage near incompatible materials (See Section V). Keep product away from children and their environment and domestic animals.

OTHER PRECAUTIONS: Special attention is drawn to the requirements of the U.S. OSHA Respirator 1910.134 should airborne exposures exceed the U.S. OSHA PEL. Inadvertent contaminants to product such as moisture, ice, snow, grease or oil can cause an explosion when charged to a molten metal bath or melting furnace (preheating metal will remove moisture from the product).

SECTION IX SPILL OR LEAK PROCEDURES

SPILL OR LEAK PROCEDURES:
1. Material in dust form-minimize exposure. Clean up using dustless methods (i.e. HEPA Vacuum). Do not use compressed air.
2. Place in closed labeled containers for recycling or disposal.
3. Keep out of waterways.

NOTE: Cleanup personnel should wear protective clothing and respiratory protection where significant dust/fume exposure exists.

OTHER PROCEDURES: For large product users or involving large product quantities, we recommend that the purchaser establish a spill prevention, control and counter measure plan. This plan should include procedures for proper storage as well as clean up of spills or leaks. The procedures should conform to safe practices and provide for proper recovery and/or disposal. Depending on the quantity spilled, notification to the U.S. National Response Center (800-424-8802) may be required in case of hazardous substances. (See USEPA and USDOT regulations: also various state and local regulations.)

WASTE DISPOSAL METHODS: May have value on a recycled basis. If disposed of, dispose of in a permitted disposal site in accordance with all federal, state and local disposal or discharge regulations.

SECTION X UNITED STATES SARA TITLE III INFORMATION

This product/mixture contains the following toxic chemical(s) subject to the reporting requirements of Section 313 of Title III of the U.S. Superfund Amendments and Reauthorization Act (SARA) of 1986 and 40 CFR Part 372. The percent by weight of each toxic chemical and its associated chemical abstract system (CAS) number are to be found in Section II of this Material Safety Data Sheet.

CHEMICAL NAME	EHS RQ (LBS) (*1)	EHS TPQ (LBS) (*2)	SEC.313 (*3)	313 CATEGORY (*4)	311-312 CATEGORIES (*5)
Copper	Not Applicable	Not Applicable	YES	Copper	H-1

-FOOTNOTES-

*1 = Reportable quantity of extremely hazardous substance, Section 302.
*2 = Threshold planning quantity, extremely hazardous substance, Section 302.
*3 = Toxic chemical list, Section 313
*4 = Chemical category as required by Section 313 (40 CFR 372.42). Subject to annual release reporting requirements.
*5 = Hazard category for SARA Section 311/312 reporting:

Health	H-1 = Immediate (ACUTE) Health Hazard	Physical	P-3 = Fire Hazard
	H-2 = Delayed (CHRONIC) Health Hazard		P-4 = Sudden Release of Pressure Hazard
			P-5 = Reactive Hazard

SECTION XI UNITED STATES CERCLA SECTION 103 INFORMATON

This product/mixture contains the following chemicals subject to the release reporting of Section 302.

CHEMICAL NAME	RQ (LBS)
Copper	5000 (*1)

-FOOTNOTES-

*1 = Reportable quantity (RQ) under CERCLA Section 302. Spills to the environment exceeding the reportable quantity in any 24-hour period must be reported to the U.S. National Response Center (800-424-8802). No reporting of releases of the hazardous substance(s) is required if the diameter of the pieces of the solid metal(s) released is equal to or exceeds 100 micrometers (0.004 inches).

SECTION XII TRANSPORTATION INFORMATION

PROPER SHIPPING NAME:	Non-regulated as shipped
TECHNICAL NAME:	N/A
HAZARD CLASS:	N/A
UN NO.:	N/A
PACKING GROUP:	N/A
EMERGENCY RESPONSE GUIDE NUMBER:	N/A
OTHER:	N/A

SECTION XIII ADDITIONAL INFORMATION

VOC CONTENT: None

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Material Safety Data Sheet

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SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME: 3M(TM) Fire Barrier Sealant FD 150+, Red
MANUFACTURER: 3M
DIVISION: Building & Commercial Services Division

ADDRESS: 3M Center
St. Paul, MN 55144-1000

EMERGENCY PHONE: 1-800-364-3577 or (651) 737-6501 (24 hours)

Issue Date: 07/14/10
Supercedes Date: 05/05/10

Document Group: 27-3508-2

Product Use:
Intended Use: Fire Protection
Specific Use: Caulk used as a passive fire protection.

SECTION 2: INGREDIENTS

Ingredient	C.A.S. No.	% by Wt
Calcium Carbonate	1317-65-3	40 - 70
Polymer	Trade Secret	10 - 30
Water	7732-18-5	5 - 10
Acrylic Emulsion	70677-00-8	5 - 10
Mineral Spirits	64742-88-7	5 - 10
Ethylene Glycol	107-21-1	1 - 5
Iron Oxide	1309-37-1	1 - 5
Plasticizer	27138-31-4	1 - 5
Ethyl hydroxyethyl cellulose	9004-58-4	0.5 - 1.5
Quartz Silica	14808-60-7	0.1 - 1

SECTION 3: HAZARDS IDENTIFICATION

3.1 EMERGENCY OVERVIEW

Specific Physical Form: Paste

Odor, Color, Grade: Red paste with low odor
General Physical Form: Solid
Immediate health, physical, and environmental hazards: Contains a chemical or chemicals which can cause cancer. May cause target organ effects.

3.2 POTENTIAL HEALTH EFFECTS

Eye Contact:
Moderate Eye Irritation: Signs/symptoms may include redness, swelling, pain, tearing, and blurred or hazy vision.

Skin Contact:
Mild Skin Irritation: Signs/symptoms may include localized redness, swelling, and itching.

Inhalation:
Respiratory Tract Irritation: Signs/symptoms may include cough, sneezing, nasal discharge, headache, hoarseness, and nose and throat pain.

May be absorbed following inhalation and cause target organ effects.

Ingestion:
Gastrointestinal Irritation: Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

May be absorbed following ingestion and cause target organ effects.

Target Organ Effects:
Cardiac Effects: Signs/symptoms may include irregular heartbeat (arrhythmia), changes in heart rate, damage to heart muscle, heart attack, and may be fatal.

Central Nervous System (CNS) Depression: Signs/symptoms may include headache, dizziness, drowsiness, incoordination, nausea, slowed reaction time, slurred speech, giddiness, and unconsciousness.

Kidney/Bladder Effects: Signs/symptoms may include changes in urine production, abdominal or lower back pain, increased protein in urine, increased blood urea nitrogen (BUN), blood in urine, and painful urination.

Respiratory Effects: Signs/symptoms may include cough, shortness of breath, chest tightness, wheezing, increased heart rate, bluish colored skin (cyanosis), sputum production, changes in lung function tests, and/or respiratory failure.

Neurological Effects: Signs/symptoms may include personality changes, lack of coordination, sensory loss, tingling or numbness of the extremities, weakness, tremors, and/or changes in blood pressure and heart rate.

Carcinogenicity:
Contains a chemical or chemicals which can cause cancer.

This product contains quartz silica. Lung cancer and silicosis have been associated with exposure to quartz silica. No user exposure to quartz silica from this product is anticipated during normal intended use. Warnings are provided to comply with US federal and state regulations.

Ingredient	C.A.S. No.	Class Description	Regulation
Quartz Silica	14808-60-7	Grp. 1: Carcinogenic to humans	International Agency for Research on Cancer

SECTION 4: FIRST AID MEASURES

4.1 FIRST AID PROCEDURES

The following first aid recommendations are based on an assumption that appropriate personal and industrial hygiene practices are followed.

Eye Contact: Flush eyes with large amounts of water. If signs/symptoms persist, get medical attention.

Skin Contact: Wash affected area with soap and water. If signs/symptoms develop, get medical attention.

Inhalation: Remove person to fresh air. If signs/symptoms develop, get medical attention.

If Swallowed: Do not induce vomiting. Give victim two glasses of water. Never give anything by mouth to an unconscious person. Get medical attention.

SECTION 5: FIRE FIGHTING MEASURES

5.1 FLAMMABLE PROPERTIES

Autoignition temperature	*Not Applicable*
Flash Point	*Not Applicable*
Flammable Limits - LEL	*Not Applicable*
Flammable Limits - UEL	*Not Applicable*
OSHA Flammability Classification:	Not Applicable

5.2 EXTINGUISHING MEDIA

Non-combustible. Choose material suitable for surrounding fire.

5.3 PROTECTION OF FIRE FIGHTERS

Special Fire Fighting Procedures: Wear full protective equipment (Bunker Gear) and a self-contained breathing apparatus (SCBA).

Unusual Fire and Explosion Hazards: Not applicable.

Note: See STABILITY AND REACTIVITY (SECTION 10) for hazardous combustion and thermal decomposition information.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal precautions

Evacuate unprotected and untrained personnel from hazard area. The spill should be cleaned up by qualified personnel. Ventilate the area with fresh air. For large spill, or spills in confined spaces, provide mechanical ventilation to disperse or exhaust vapors, in accordance with good industrial hygiene practice. Warning! A motor could be an ignition source and could cause flammable gases or vapors in the spill area to burn or explode.

Environmental procedures
Place in a closed container approved for transportation by appropriate authorities. Dispose of collected material as soon as possible.

Clean-up methods
Observe precautions from other sections. Call 3M- HELPS line (1-800-364-3577) for more information on handling and managing the spill. Collect as much of the spilled material as possible. Clean up residue.

In the event of a release of this material, the user should determine if the release qualifies as reportable according to local, state, and federal regulations.

SECTION 7: HANDLING AND STORAGE

7.1 HANDLING
Keep out of the reach of children. Avoid breathing of vapors. Avoid eye contact. Do not eat, drink or smoke when using this product. Wash exposed areas thoroughly with soap and water. Avoid contact with oxidizing agents.

7.2 STORAGE
Store away from oxidizing agents. Store in a cool, dry place.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 ENGINEERING CONTROLS
Use in a well-ventilated area.

8.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

8.2.1 Eye/Face Protection
Avoid eye contact.
The following eye protection(s) are recommended: Safety Glasses with side shields
Indirect Vented Goggles
.

8.2.2 Skin Protection
Avoid skin contact.

Select and use gloves and/or protective clothing to prevent skin contact based on the results of an exposure assessment. Consult with your glove and/or protective clothing manufacturer for selection of appropriate compatible materials.
Gloves made from the following material(s) are recommended: Butyl Rubber
Neoprene
.

8.2.3 Respiratory Protection
Avoid breathing of vapors. Under normal use conditions, airborne exposures are not expected to be significant enough to require respiratory protection.
Select one of the following NIOSH approved respirators based on airborne concentration of contaminants and in accordance with

OSHA regulations: Half facepiece or fullface air-purifying respirator with organic vapor cartridges
. Consult the current 3M Respiratory Selection Guide for additional information or call 1-800-243-4630 for 3M technical assistance.

8.2.4 Prevention of Swallowing
Do not eat, drink or smoke when using this product. Wash exposed areas thoroughly with soap and water.

8.3 EXPOSURE GUIDELINES

Ingredient	Authority	Type	Limit	Additional Information
Ethylene Glycol	ACGIH	CEIL, as aerosol	100 mg/m3	
Ethylene Glycol	CMRG	CEIL, as vapor and aerosol	100 mg/m3	
Iron Oxide	ACGIH	TWA, respirable fraction	5 mg/m3	
Iron Oxide	OSHA	TWA, as fume	10 mg/m3	
Calcium Carbonate	OSHA	TWA, respirable fraction	5 mg/m3	
Calcium Carbonate	OSHA	TWA, as total dust	15 mg/m3	
Mineral Spirits	CMRG	TWA	100 ppm	
Quartz Silica	ACGIH	TWA, respirable fraction	0.025 mg/m3	
Quartz Silica	OSHA	TWA concentration, respirable	0.1 mg/m3	
Quartz Silica	OSHA	TWA concentration, as total dust	0.3 mg/m3	

SOURCE OF EXPOSURE LIMIT DATA:
ACGIH: American Conference of Governmental Industrial Hygienists
CMRG: Chemical Manufacturer Recommended Guideline
OSHA: Occupational Safety and Health Administration
AIHA: American Industrial Hygiene Association Workplace Environmental Exposure Level (WEEL)

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Specific Physical Form:	Paste
Odor, Color, Grade:	Red paste with low odor
General Physical Form:	Solid
Autoignition temperature	*Not Applicable*
Flash Point	*Not Applicable*
Flammable Limits - LEL	*Not Applicable*
Flammable Limits - UEL	*Not Applicable*

Specific Gravity	1.45 [*Ref Std:* WATER=1]
Melting point	*No Data Available*
Volatile Organic Compounds	< 15 % weight
VOC Less H2O & Exempt Solvents	< 250 g/l

SECTION 10: STABILITY AND REACTIVITY

Stability: Stable.

Materials and Conditions to Avoid:
10.1 Conditions to avoid
None known

10.2 Materials to avoid
Strong oxidizing agents

Hazardous Polymerization: Hazardous polymerization will not occur.

Hazardous Decomposition or By-Products

Substance	Condition
Carbon monoxide	Not Specified
Carbon dioxide	Not Specified

SECTION 11: TOXICOLOGICAL INFORMATION

Please contact the address listed on the first page of the MSDS for Toxicological Information on this material and/or its components.

SECTION 12: ECOLOGICAL INFORMATION

ECOTOXICOLOGICAL INFORMATION

Not determined.

CHEMICAL FATE INFORMATION

Not determined.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Method: For quantities <100 lbs. (50kg): dispose of waste product in a sanitary landfill. For larger quantities: Dispose of waste product in a facility permitted to accept chemical waste.

EPA Hazardous Waste Number (RCRA): Not regulated

Since regulations vary, consult applicable regulations or authorities before disposal.

SECTION 14:TRANSPORT INFORMATION

ID Number(s):
98-0400-5598-4, 98-0400-5599-2, 98-0400-5600-8, 98-0400-5601-6

Please contact the emergency numbers listed on the first page of the MSDS for Transportation Information for this material.

SECTION 15: REGULATORY INFORMATION

US FEDERAL REGULATIONS
Contact 3M for more information.

311/312 Hazard Categories:
Fire Hazard - No Pressure Hazard - No Reactivity Hazard - No Immediate Hazard - Yes Delayed Hazard - Yes

Section 313 Toxic Chemicals subject to the reporting requirements of that section and 40 CFR part 372 (EPCRA):

Ingredient	C.A.S. No	% by Wt
Ethylene Glycol	107-21-1	1 - 5

STATE REGULATIONS
Contact 3M for more information.

CHEMICAL INVENTORIES
The components of this product are in compliance with the chemical notification requirements of TSCA.

The components of this product are listed on the Canadian Domestic Substances List.

Contact 3M for more information.

INTERNATIONAL REGULATIONS
Contact 3M for more information.

This MSDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.

SECTION 16: OTHER INFORMATION

NFPA Hazard Classification

Health: 2 **Flammability:** 0 **Reactivity:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

HMIS Hazard Classification

Health: 2 **Flammability:** 0 **Reactivity:** 0 **Protection:** X - See PPE section.

Hazardous Material Identification System (HMIS(r)) hazard ratings are designed to inform employees of chemical hazards in the workplace. These ratings are based on the inherent properties of the material under expected conditions of normal use and are not intended for use in emergency situations. HMIS(r) ratings are to be used with a fully implemented HMIS(r) program. HMIS(r) is a registered mark of the National Paint and Coatings Association (NPCA).

Reason for Reissue: Added new sku

Revision Changes:
Section 3: Carcinogenicity comment was modified.
Section 2: Ingredient table was modified.
Section 6: Environmental procedures heading was added.
Section 6: Personal precautions heading was added.
Section 6: Clean-up methods heading was added.
Section 6: Release measures heading was deleted.

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SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME: 3M FireBarrier™ Sealant IC 15 WB+
MANUFACTURER: 3M
DIVISION: Building & Commercial Services Division

ADDRESS: 3M Center, St. Paul, MN 55144-1000

EMERGENCY PHONE: 1-800-364-3577 or (651) 737-6501 (24 hours)

Issue Date: 07/11/12
Supercedes Date: 09/16/11

Document Group: 19-9776-6

Product Use:
Specific Use: Fire Barrier Sealant
Intended Use: Sealant

SECTION 2: INGREDIENTS

Ingredient	C.A.S. No.	% by Wt
Calcium Carbonate	1317-65-3	30 - 60
Water	7732-18-5	10 - 30
Polymer	Trade Secret	10 - 30
ZINC BORATE 2335	138265-88-0	3 - 7
Sodium Silicate	1344-09-8	3 - 7
Fiberglass	65997-17-3	0.5 - 1.5
POLYOXYETHYLENE MONOOCTYLPHENYL ETHER	9036-19-5	0.1 - 1
SODIUM NONYLPHENYL POLYETHOXY ETHER SULFATE	9014-90-8	< 1
Quartz Silica	14808-60-7	< 0.5

SECTION 3: HAZARDS IDENTIFICATION

3.1 EMERGENCY OVERVIEW

Specific Physical Form: Paste
Odor, Color, Grade: Light yellow viscous paste with a mild odor
General Physical Form: Solid

Immediate health, physical, and environmental hazards: May cause target organ effects. Contains a chemical or chemicals which can cause cancer.

3.2 POTENTIAL HEALTH EFFECTS

Eye Contact:
Moderate Eye Irritation: Signs/symptoms may include redness, swelling, pain, tearing, and blurred or hazy vision.

Skin Contact:
Mild Skin Irritation: Signs/symptoms may include localized redness, swelling, and itching.

Inhalation:
Respiratory Tract Irritation: Signs/symptoms may include cough, sneezing, nasal discharge, headache, hoarseness, and nose and throat pain.

Ingestion:
Gastrointestinal Irritation: Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

May be absorbed following ingestion and cause target organ effects.

Target Organ Effects:
Prolonged or repeated exposure by ingestion may cause:
Kidney/Bladder Effects: Signs/symptoms may include changes in urine production, abdominal or lower back pain, increased protein in urine, increased blood urea nitrogen (BUN), blood in urine, and painful urination.

Carcinogenicity:
Contains a chemical or chemicals which can cause cancer.

Ingredient	C.A.S. No.	Class Description	Regulation
Quartz Silica	14808-60-7	Grp. 1: Carcinogenic to humans	International Agency for Research on Cancer

SECTION 4: FIRST AID MEASURES

4.1 FIRST AID PROCEDURES

The following first aid recommendations are based on an assumption that appropriate personal and industrial hygiene practices are followed.

Eye Contact: Immediately flush eyes with large amounts of water for at least 15 minutes. Get immediate medical attention.
Skin Contact: Wash affected area with soap and water. If signs/symptoms develop, get medical attention.
Inhalation: Remove person to fresh air. If signs/symptoms develop, get medical attention.
If Swallowed: Do not induce vomiting unless instructed to do so by medical personnel. Give victim two glasses of water. Never give anything by mouth to an unconscious person. Get medical attention.

SECTION 5: FIRE FIGHTING MEASURES

5.1 FLAMMABLE PROPERTIES

Autoignition temperature	*No Data Available*
Flash Point	Flash point > 93 °C (200 °F)
Flammable Limits(LEL)	*Not Applicable*
Flammable Limits(UEL)	*Not Applicable*

OSHA Flammability Classification: Not Applicable

5.2 EXTINGUISHING MEDIA
Non-combustible. Choose material suitable for surrounding fire.

5.3 PROTECTION OF FIRE FIGHTERS

Special Fire Fighting Procedures: Wear full protective equipment (Bunker Gear) and a self-contained breathing apparatus (SCBA).

Unusual Fire and Explosion Hazards: No unusual fire or explosion hazards are anticipated.

Note: See STABILITY AND REACTIVITY (SECTION 10) for hazardous combustion and thermal decomposition information.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures
Evacuate unprotected and untrained personnel from hazard area. The spill should be cleaned up by qualified personnel. Ventilate the area with fresh air. For large spill, or spills in confined spaces, provide mechanical ventilation to disperse or exhaust vapors, in accordance with good industrial hygiene practice. Warning! A motor could be an ignition source and could cause flammable gases or vapors in the spill area to burn or explode.

6.2. Environmental precautions
Place in a closed container approved for transportation by appropriate authorities. Dispose of collected material as soon as possible.

Clean-up methods
Observe precautions from other sections. Call 3M- HELPS line (1-800-364-3577) for more information on handling and managing the spill. Collect as much of the spilled material as possible. Clean up residue.

In the event of a release of this material, the user should determine if the release qualifies as reportable according to local, state, and federal regulations.

SECTION 7: HANDLING AND STORAGE

7.1 HANDLING
Keep out of the reach of children. For industrial or professional use only. Avoid eye contact with dust or airborne particles. Do not eat, drink or smoke when using this product. Wash exposed areas thoroughly with soap and water.

7.2 STORAGE
Store under normal warehouse conditions.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 ENGINEERING CONTROLS
Use in a well-ventilated area.

8.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

8.2.1 Eye/Face Protection

Avoid eye contact.
The following eye protection(s) are recommended: Safety Glasses with side shields
Indirect Vented Goggles
.

8.2.2 Skin Protection
Avoid skin contact.

Select and use gloves and/or protective clothing to prevent skin contact based on the results of an exposure assessment. Consult with your glove and/or protective clothing manufacturer for selection of appropriate compatible materials.
Gloves made from the following material(s) are recommended: Butyl Rubber
Nitrile Rubber
.

8.2.3 Respiratory Protection
Under normal use conditions, airborne exposures are not expected to be significant enough to require respiratory protection.

8.2.4 Prevention of Swallowing
Do not eat, drink or smoke when using this product. Wash exposed areas thoroughly with soap and water.

8.3 EXPOSURE GUIDELINES

Ingredient	Authority	Type	Limit	Additional Information
Calcium Carbonate	OSHA	TWA, respirable fraction	5 mg/m3	
Calcium Carbonate	OSHA	TWA, as total dust	15 mg/m3	
Fiberglass	Manufacturer determined	TWA, as dust	10 mg/m3	
Quartz Silica	ACGIH	TWA, respirable fraction	0.025 mg/m3	
Quartz Silica	OSHA	TWA concentration, respirable	0.1 mg/m3	
Quartz Silica	OSHA	TWA concentration, as total dust	0.3 mg/m3	

SOURCE OF EXPOSURE LIMIT DATA:
ACGIH: American Conference of Governmental Industrial Hygienists
CMRG: Chemical Manufacturer Recommended Guideline
OSHA: Occupational Safety and Health Administration
AIHA: American Industrial Hygiene Association Workplace Environmental Exposure Level (WEEL)

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Specific Physical Form:	Paste
Odor, Color, Grade:	Light yellow viscous paste with a mild odor
General Physical Form:	Solid
Autoignition temperature	*No Data Available*
Flash Point	Flash point > 93 °C (200 °F)
Flammable Limits(LEL)	*Not Applicable*
Flammable Limits(UEL)	*Not Applicable*
Boiling Point	*Not Applicable*

Specific Gravity	1.4 [*Ref Std:* WATER=1]
Melting point	*No Data Available*
Solubility in Water	Moderate
Volatile Organic Compounds	< 2 g/l
VOC Less H2O & Exempt Solvents	< 2 g/l

SECTION 10: STABILITY AND REACTIVITY

Stability: Stable.

Materials and Conditions to Avoid:
10.1 Conditions to avoid
None known

10.2 Materials to avoid
None known

Hazardous Polymerization: Hazardous polymerization will not occur.

Hazardous Decomposition or By-Products

Substance	Condition
Carbon monoxide	Not Specified
Carbon dioxide	Not Specified

SECTION 11: TOXICOLOGICAL INFORMATION

Please contact the address listed on the first page of the MSDS for Toxicological Information on this material and/or its components.

SECTION 12: ECOLOGICAL INFORMATION

ECOTOXICOLOGICAL INFORMATION

Not determined.

CHEMICAL FATE INFORMATION

Not determined.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Method: For quantities <100 lbs. (50kg): dispose of waste product in a sanitary landfill. For larger quantities: Dispose of waste product in a facility permitted to accept chemical waste.

EPA Hazardous Waste Number (RCRA): Not regulated

Since regulations vary, consult applicable regulations or authorities before disposal.

SECTION 14:TRANSPORT INFORMATION

ID Number(s):
42-0016-4768-6, 42-0016-4769-4, 42-0016-4770-2, 98-0400-5509-1, 98-0400-5510-9, 98-0400-5511-7, 98-0400-5512-5, 98-0400-5630-5

For Transport Information, please visit http://3M.com/Transportinfo or call 1-800-364-3577 or 651-737-6501.

SECTION 15: REGULATORY INFORMATION

US FEDERAL REGULATIONS

Contact 3M for more information.

Section 313 Toxic Chemicals subject to the reporting requirements of that section and 40 CFR part 372 (EPCRA):

Ingredient	C.A.S. No	% by Wt
ZINC BORATE 2335 (ZINC COMPOUNDS)	138265-88-0	3 - 7

STATE REGULATIONS

Contact 3M for more information.

CALIFORNIA PROPOSITION 65

Ingredient	C.A.S. No.	Classification
SILICA, CRYSTALLINE (AIRBORNE PARTICLES OF RESPIRABLE SIZE)	SEQ677	**Carcinogen

** WARNING: contains a chemical which can cause cancer.

CHEMICAL INVENTORIES

The components of this product are in compliance with the chemical notification requirements of TSCA.

All applicable chemical ingredients in this material are listed on the European Inventory of Existing Chemical Substances (EINECS), or are exempt polymers whose monomers are listed on EINECS. This product complies with the New Zealand Hazardous Substances and New Organisms Act (1996).

Contact 3M for more information.

INTERNATIONAL REGULATIONS

Contact 3M for more information.

This MSDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.

SECTION 16: OTHER INFORMATION

NFPA Hazard Classification

Health: 2 **Flammability:** 0 **Reactivity:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

Revision Changes:
Section 1: Product name was modified.
Section 3: Potential effects from skin contact information was modified.
Section 3: Potential effects from ingestion information was modified.
Section 7: Handling information was modified.
Section 4: First aid for skin contact - decontamination - was modified.
Section 4: First aid for skin contact - medical assistance - was modified.
Section 3: Immediate other hazard(s) was modified.
Page Heading: Product name was modified.
Section 15: Inventories information was modified.
Section 5: Flash point information was modified.
Section 9: Flash point information was modified.
Section 14: ID Number(s) Template 1 was modified.
Section 2: Ingredient table was modified.
Section 8: Exposure guidelines ingredient information was modified.
Section 6: Personal precautions information was modified.
Copyright was modified.
Section 3: Other potential health effects heading was added.
Section 3: Other health effects information was added.
Section 15: California proposition 65 ingredient information was added.
Section 15: California proposition 65 heading was added.
Section 15: California proposition 65 cancer warning was added.
Section 3: Carcinogenicity comment was deleted.
Section 4: First aid for skin contact - termination of exposure - was deleted.
Section 4: First aid for skin contact - handling - was deleted.

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Material Safety Data Sheet

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This material safety data sheet (MSDS) is provided as a courtesy in response to a customer request. This product is not regulated under, and a MSDS is not required for this product by the OSHA Hazard Communication Standard (29 CFR 1910.1200) because, when used as recommended or under ordinary conditions, it should not present a health and safety hazard. However, use or processing of the product not in accordance with the product's recommendations or not under ordinary conditions may affect the performance of the product and may present potential health and safety hazards.

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME: Tuck In Wrap Strips
MANUFACTURER: 3M
DIVISION: Building & Commercial Services Division

ADDRESS: 3M Center, St. Paul, MN 55144-1000

EMERGENCY PHONE: 1-800-364-3577 or (651) 737-6501 (24 hours)

Issue Date: 05/17/11
Supercedes Date: 07/01/10

Document Group: 28-4146-8

Product Use:
Intended Use: Fire Protection

SECTION 2: INGREDIENTS

Ingredient	C.A.S. No.	% by Wt
Graphite	7782-42-5	40 - 70
Synthetic Rubber	None	15 - 40
Alumina Trihydrate	21645-51-2	5 - 10
Plasticizer	26761-40-0	1 - 5
Color	None	< 1
Antioxidant	None	< 1

SECTION 3: HAZARDS IDENTIFICATION

3.1 EMERGENCY OVERVIEW

Odor, Color, Grade: Rubber composite with slight odor

General Physical Form: Solid
Immediate health, physical, and environmental hazards: This product, when used under reasonable conditions and in accordance with the 3M directions for use, should not present a health hazard. However, use or processing of the product in a manner not in accordance with the product's directions for use may affect the performance of the product and may present potential health and safety hazards.

3.2 POTENTIAL HEALTH EFFECTS

Eye Contact:
No health effects are expected.

Skin Contact:
No health effects are expected.

Inhalation:
No health effects are expected.

Ingestion:
No health effects are expected.

SECTION 4: FIRST AID MEASURES

4.1 FIRST AID PROCEDURES

The following first aid recommendations are based on an assumption that appropriate personal and industrial hygiene practices are followed.

Eye Contact: No need for first aid is anticipated.
Skin Contact: No need for first aid is anticipated.
Inhalation: No need for first aid is anticipated.
If Swallowed: No need for first aid is anticipated.

SECTION 5: FIRE FIGHTING MEASURES

5.1 FLAMMABLE PROPERTIES

Flash Point	*No Data Available*
Flammable Limits(LEL)	*Not Applicable*
Flammable Limits(UEL)	*Not Applicable*

5.2 EXTINGUISHING MEDIA
Use fire extinguishers with class B extinguishing agents (e.g., dry chemical, carbon dioxide).

5.3 PROTECTION OF FIRE FIGHTERS

Special Fire Fighting Procedures: Wear full protective equipment (Bunker Gear) and a self-contained breathing apparatus (SCBA).

Unusual Fire and Explosion Hazards: No unusual fire or explosion hazards are anticipated.

Note: See STABILITY AND REACTIVITY (SECTION 10) for hazardous combustion and thermal decomposition information.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures
Not applicable.

6.2. Environmental precautions
Not applicable.

Clean-up methods
Not applicable.

SECTION 7: HANDLING AND STORAGE

7.1 HANDLING
This product is considered to be an article which does not release or otherwise result in exposure to a hazardous chemical under normal use conditions.

7.2 STORAGE
Store under normal warehouse conditions.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 ENGINEERING CONTROLS
Not applicable.

8.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

8.2.1 Eye/Face Protection
Not applicable.

8.2.2 Skin Protection
Not applicable.

8.2.3 Respiratory Protection
Under normal use conditions, airborne exposures are not expected to be significant enough to require respiratory protection.

8.2.4 Prevention of Swallowing
Not applicable.

8.3 EXPOSURE GUIDELINES

Ingredient	Authority	Type	Limit	Additional Information
Graphite	ACGIH	TWA, respirable fraction	2 mg/m3	
Graphite	OSHA	TWA	15 millions of particles/cu. ft.	

SOURCE OF EXPOSURE LIMIT DATA:
ACGIH: American Conference of Governmental Industrial Hygienists
CMRG: Chemical Manufacturer Recommended Guideline
OSHA: Occupational Safety and Health Administration
AIHA: American Industrial Hygiene Association Workplace Environmental Exposure Level (WEEL)

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Odor, Color, Grade:	Rubber composite with slight odor
General Physical Form:	Solid
Flash Point	*No Data Available*
Flammable Limits(LEL)	*Not Applicable*
Flammable Limits(UEL)	*Not Applicable*
Density	1.6 g/ml
Specific Gravity	1.6
Solubility in Water	Nil

SECTION 10: STABILITY AND REACTIVITY

Stability: Stable.

Materials and Conditions to Avoid:
10.1 Conditions to avoid
None known

10.2 Materials to avoid
None known

Hazardous Polymerization: Hazardous polymerization will not occur.

Hazardous Decomposition or By-Products

Substance	Condition
Carbon monoxide	Not Specified
Carbon dioxide	Not Specified

Hazardous Decomposition: Under recommended usage conditions, hazardous decomposition products are not expected. Hazardous decomposition products may occur as a result of oxidation, heating, or reaction with another material.
Sulfuric acid and nitric acid may be generated during product decomposition.

SECTION 11: TOXICOLOGICAL INFORMATION

Please contact the address listed on the first page of the MSDS for Toxicological Information on this material and/or its components.

SECTION 12: ECOLOGICAL INFORMATION

ECOTOXICOLOGICAL INFORMATION

Not applicable.

CHEMICAL FATE INFORMATION

Not applicable.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Method: Dispose of waste product in a sanitary landfill. As a disposal alternative, incinerate in an industrial or commercial facility in the presence of a combustible material.

EPA Hazardous Waste Number (RCRA): Not regulated

Since regulations vary, consult applicable regulations or authorities before disposal.

SECTION 14:TRANSPORT INFORMATION

ID Number(s):
98-0400-5602-4, 98-0400-5603-2, 98-0400-5605-7, 98-0400-5606-5

For Transport Information, please visit http://3M.com/Transportinfo or call 1-800-364-3577 or 651-737-6501.

SECTION 15: REGULATORY INFORMATION

US FEDERAL REGULATIONS
Contact 3M for more information.

STATE REGULATIONS
Contact 3M for more information.

CHEMICAL INVENTORIES
This product is an article as defined by TSCA regulations, and is exempt from TSCA Inventory listing requirements.

Contact 3M for more information.

INTERNATIONAL REGULATIONS
Contact 3M for more information.

This MSDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.

SECTION 16: OTHER INFORMATION

Revision Changes:
Section 16: Disclaimer (second paragraph) was modified.
Section 14: Transportation legal text was modified.
Section 5: Flammable limits (UE) information was modified.
Section 5: Flammable limits (LEL) information was modified.
Section 9: Flammable limits (LEL) information was modified.
Section 9: Flammable limits (UEL) information was modified.
Section 2: Ingredient table was modified.
Section 8: Exposure guidelines ingredient information was modified.
Section 6: 6.2. Environmental precautions heading was modified.
Section 6: 6.1. Personal precautions, protective equipment and emergency procedures heading was modified.
Section 10: Hazardous decomposition or by-products comment was added.
Section 16: Web address was added.
Section 1: Address was added.
Copyright was added.
Company logo was added.
Telephone header was added.
Company Telephone was added.
Section 1: Emergency phone information was added.
Section 1: Emergency phone information was deleted.
Company Logo was deleted.
Copyright was deleted.
Section 16: Web address heading was deleted.
Section 1: Address line 1 was deleted.
Section 1: Address line 2 was deleted.

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